

Exotic Signals from Icy Worlds

A collection of seismic data that you cannot collect above 0°C



Andreas Fichtner

ETH – Seismology & Wave Physics

The cryosphere is much more **dynamic** than the lithosphere [over human time scales].

The cryosphere is much more **dynamic** than the lithosphere [over human time scales].

Cryospheric processes → wide spectrum of **seismic signals** [not used to this in classical seismology].

The cryosphere is much more **dynamic** than the lithosphere [over human time scales].

Cryospheric processes → wide spectrum of **seismic signals** [not used to this in classical seismology].

Many of these processes have concrete **societal relevance**.

The cryosphere is much more **dynamic** than the lithosphere [over human time scales].

Cryospheric processes → wide spectrum of **seismic signals** [not used to this in classical seismology].

Many of these processes have concrete **societal relevance**.

New **technologies** [e.g., fibre-optic sensing] → unprecedented spatio-temporal **resolution**.

The cryosphere is much more **dynamic** than the lithosphere [over human time scales].

Cryospheric processes → wide spectrum of **seismic signals** [not used to this in classical seismology].

Many of these processes have concrete **societal relevance**.

New **technologies** [e.g., fibre-optic sensing] → unprecedented spatio-temporal **resolution**.

These are very exciting times for cryoseismology!

Talk structure

Part I

A brief survey of cryospheric DAS signals
and what we may learn from them about the underlying processes

Talk structure

Part I

A brief survey of cryospheric DAS signals
and what we may learn from them about the underlying processes

Part II

A closer look

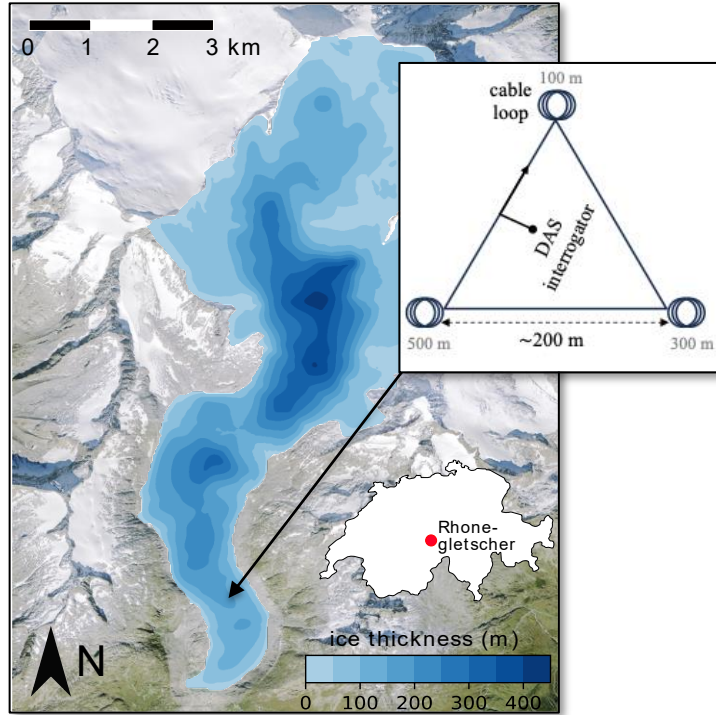
- 1: Floating ice sheet resonance
- 2: Wave guiding in firn layers
- 3: Seismic cascades in ice streams

Part I

A brief survey of cryospheric DAS signals
and what we may learn from them about the underlying processes

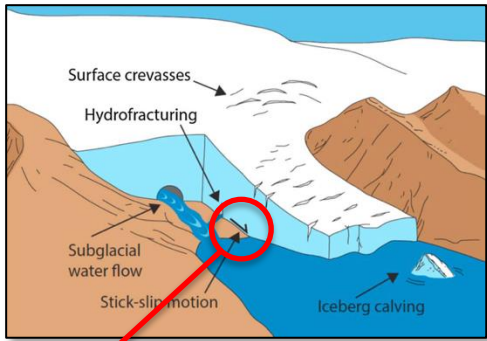
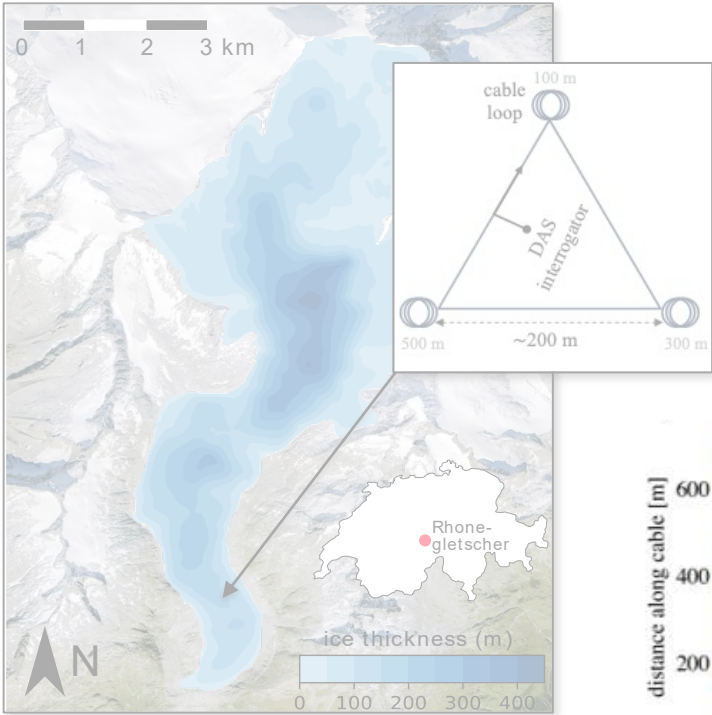
GLACIAL QUAKES [Fabian Walter, Patrick Paitz, Dominik Gräff, Andreas Fichtner, et al.]

GLACIAL QUAKES [Fabian Walter, Patrick Paitz, Dominik Gräff, Andreas Fichtner, et al.]

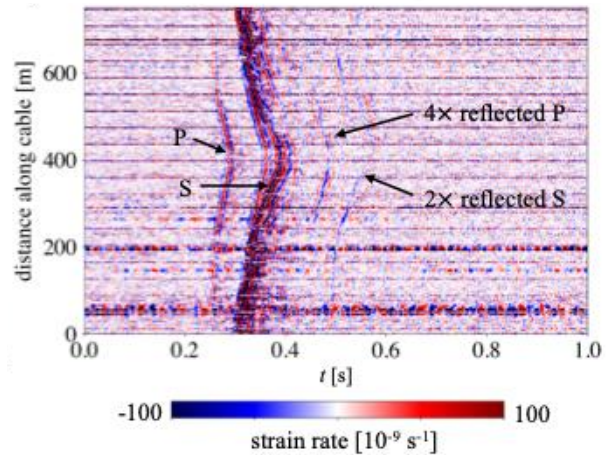


GLACIAL QUAKE

[Fabian Walter, Patrick Paitz, Dominik Gräff, Andreas Fichtner, et al.]

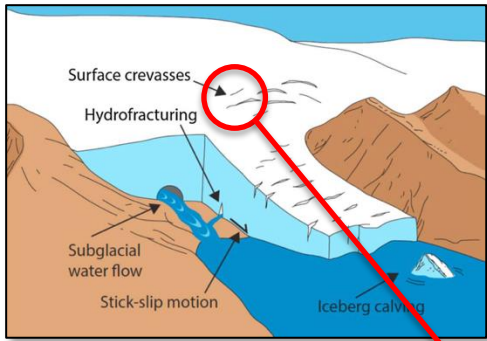
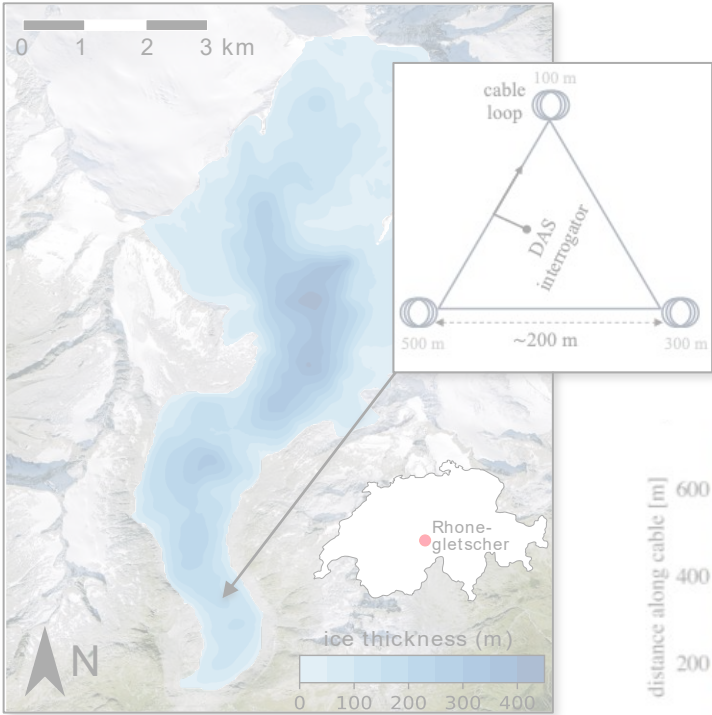


Podolskiy & Walter, 2016

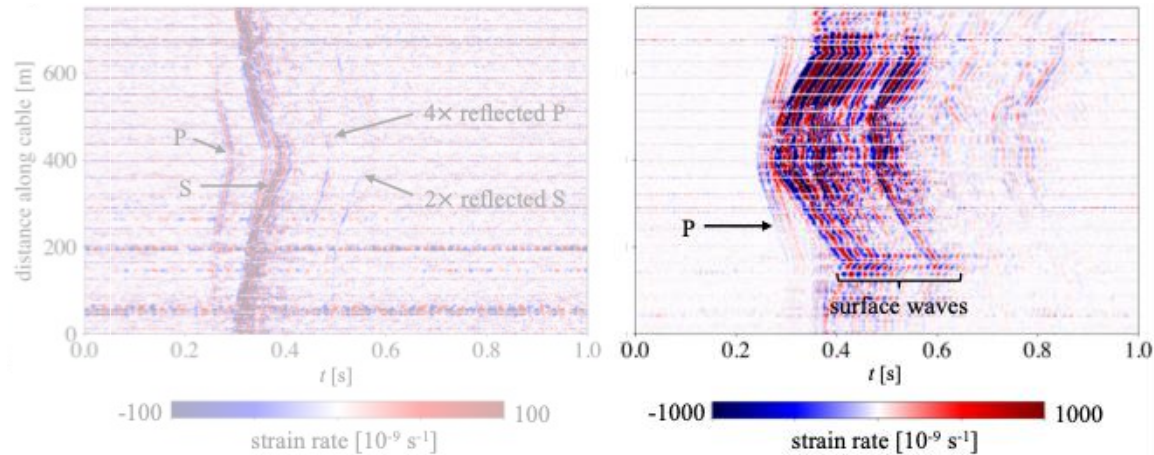


GLACIAL QUAKE

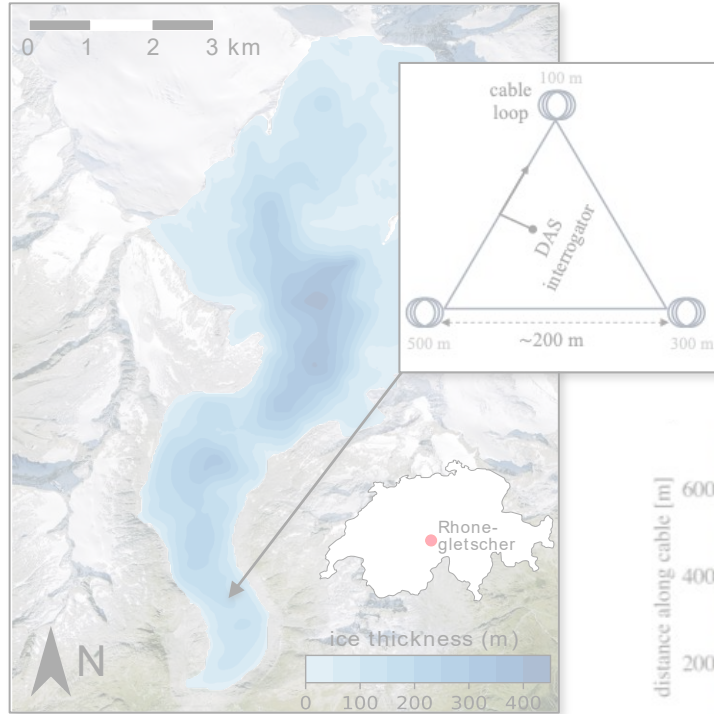
[Fabian Walter, Patrick Paitz, Dominik Gräff, Andreas Fichtner, et al.]



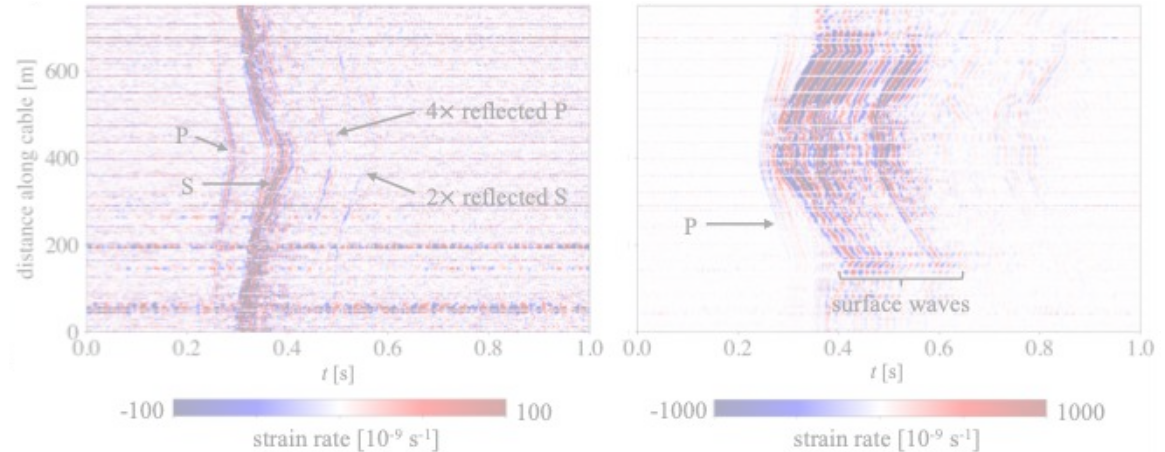
Podolskiy & Walter, 2016



GLACIAL QUAKES [Fabian Walter, Patrick Paitz, Dominik Gräff, Andreas Fichtner, et al.]



- Brittle contribution to glacial 'flow'
- Glacial dynamics
- Link to sea-level rise



LAKE ICE QUAKES [Cédric Schmelzbach, Ariana David, Pascal Edme, Ariane Lanteri]

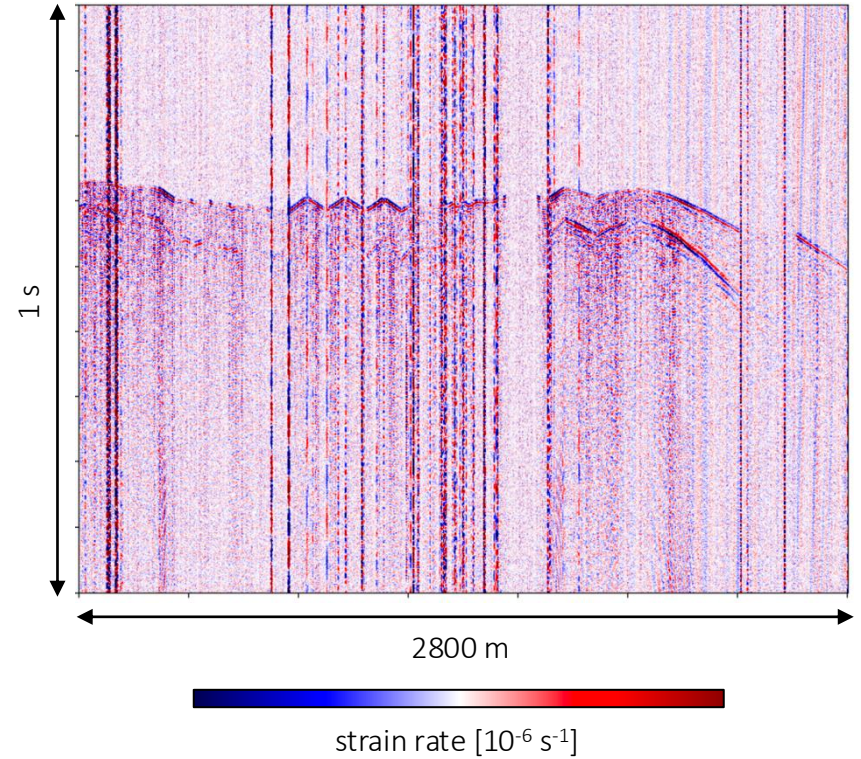
LAKE ICE QUAKES [Cédric Schmelzbach, Ariana David, Pascal Edme, Ariane Lanteri]



Lake St. Moritz in winter



LAKE ICE QUAKES [Cédric Schmelzbach, Ariana David, Pascal Edme, Ariane Lanteri]

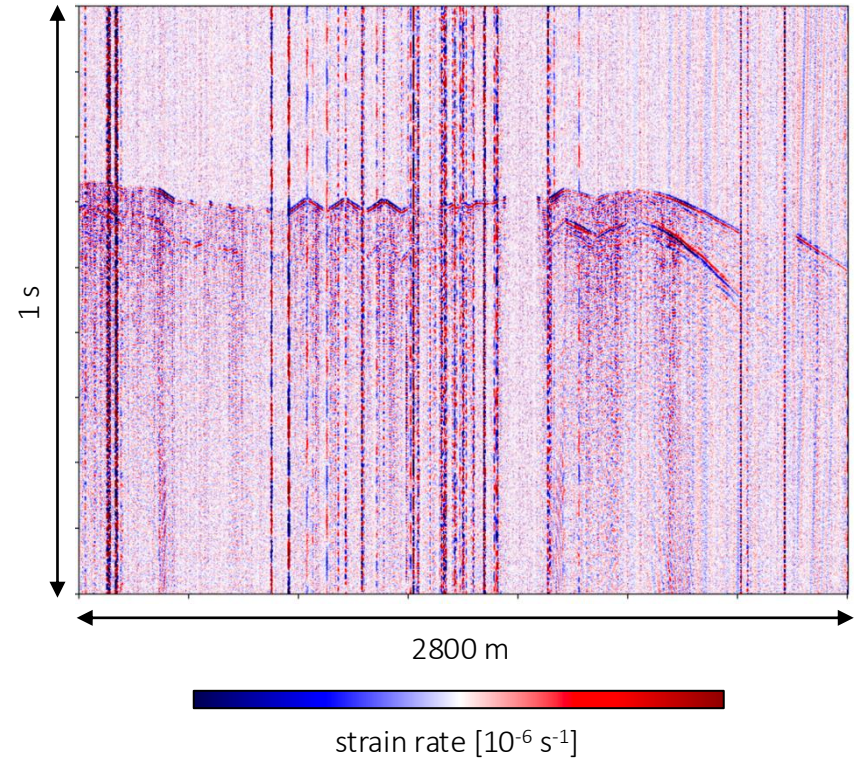


LAKE ICE QUAKES [Cédric Schmelzbach, Ariana David, Pascal Edme, Ariane Lanteri]



- Spatially variable ice thickness
- Ice stability
- Potential to monitor ice highways

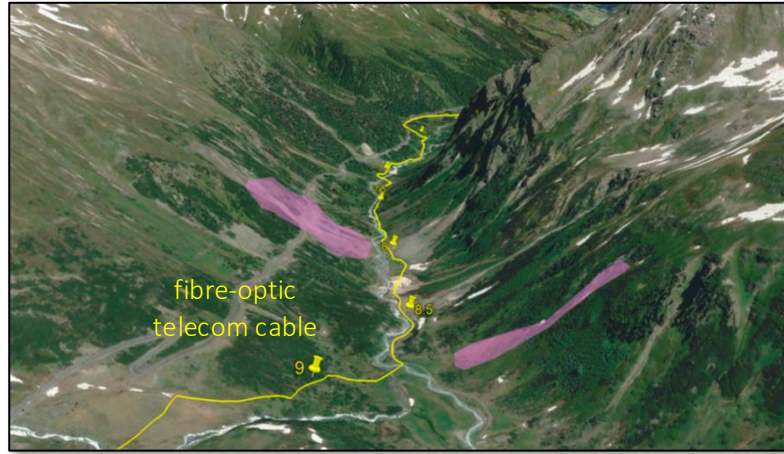
See talk by Cédric Schmelzbach!



AVALANCHES

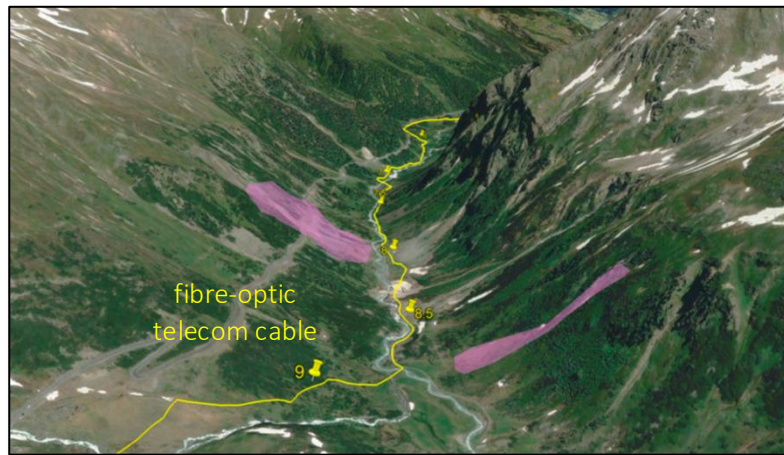
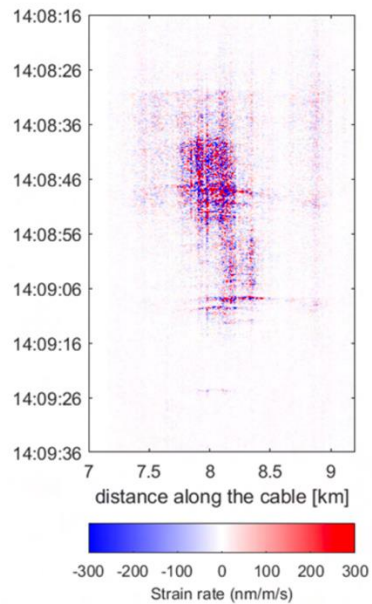
[Pascal Edme, Alec van Herwijnen, Andreas Fichtner et al.]

AVALANCHES [Pascal Edme, Alec van Herwijnen, Andreas Fichtner et al.]

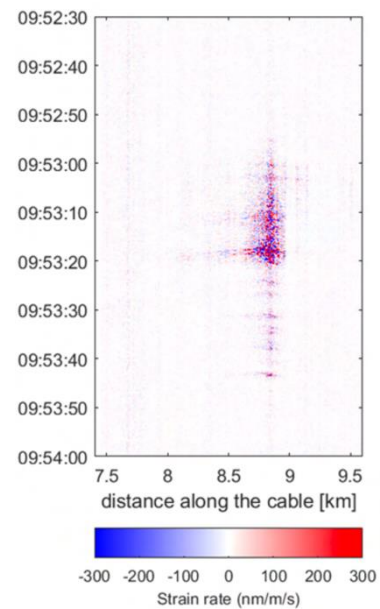


Fluela pass road, Swiss Alps

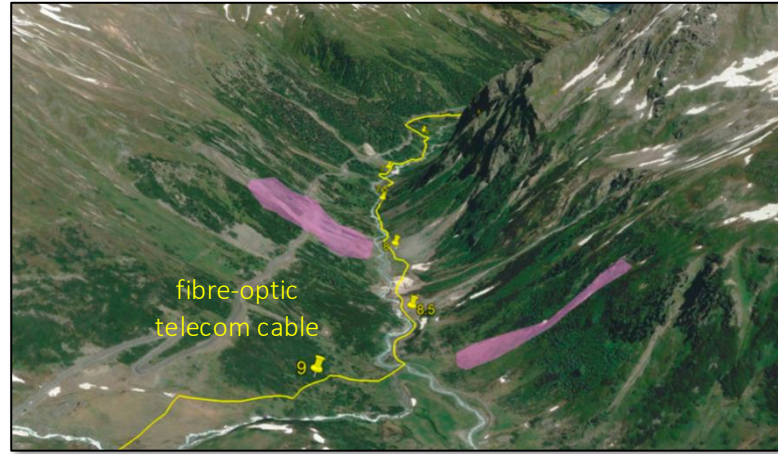
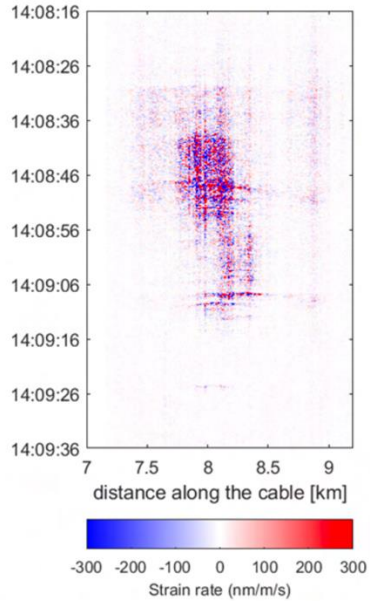
AVALANCHES [Pascal Edme, Alec van Herwijnen, Andreas Fichtner et al.]



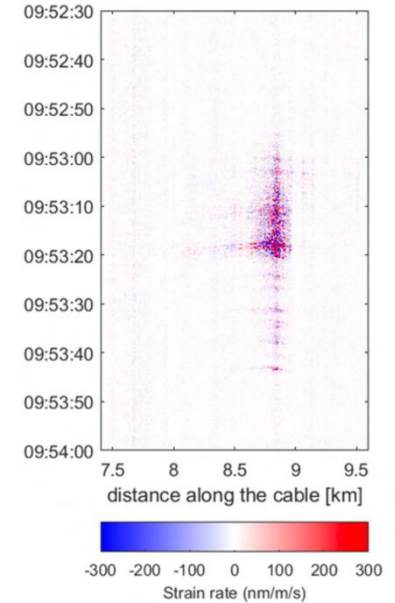
Fluela pass road, Swiss Alps



AVALANCHES [Pascal Edme, Alec van Herwijnen, Andreas Fichtner et al.]



Fluela pass road, Swiss Alps



- Low-cost, continuous avalanche monitoring
- Combine with traffic monitoring
- Close link to landslide monitoring

WHUMPFS [Janos Benz, Andreas Fichtner, Coen Hofstede et al.]

WHUMPFS [Janos Benz, Andreas Fichtner, Coen Hofstede et al.]

Depth hoar layer



Jamieson & Schweizer, 2000

Slab cracking atop depth hoar layer

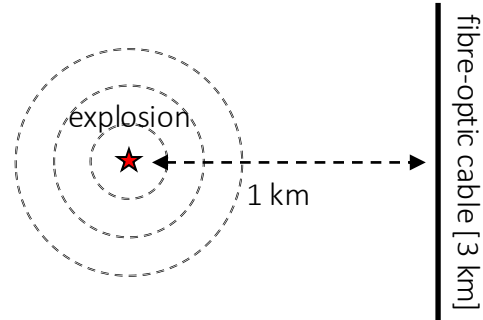


Knoff, 2016

Depth hoar layer



Jamieson & Schweizer, 2000



Slab cracking atop depth hoar layer



Knoff, 2016

WHUMPFS [Janos Benz, Andreas Fichtner, Coen Hofstede et al.]

Depth hoar layer

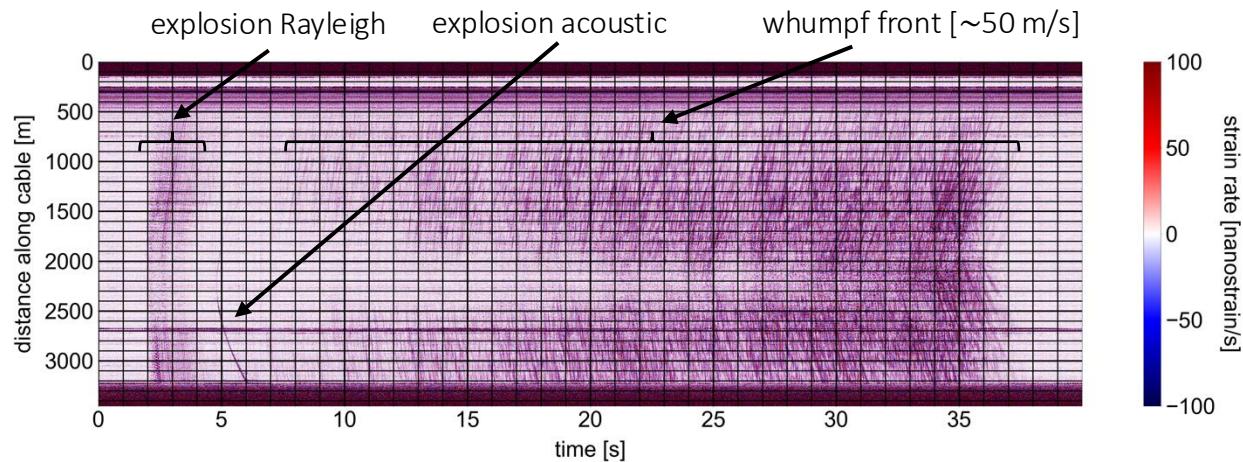
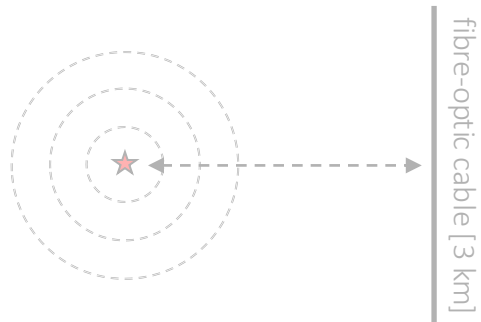


Jamieson & Schweizer, 2000

Slab cracking atop depth hoar layer



Knoff, 2016



WHUMPFS [Janos Benz, Andreas Fichtner, Coen Hofstede et al.]

Depth hoar layer

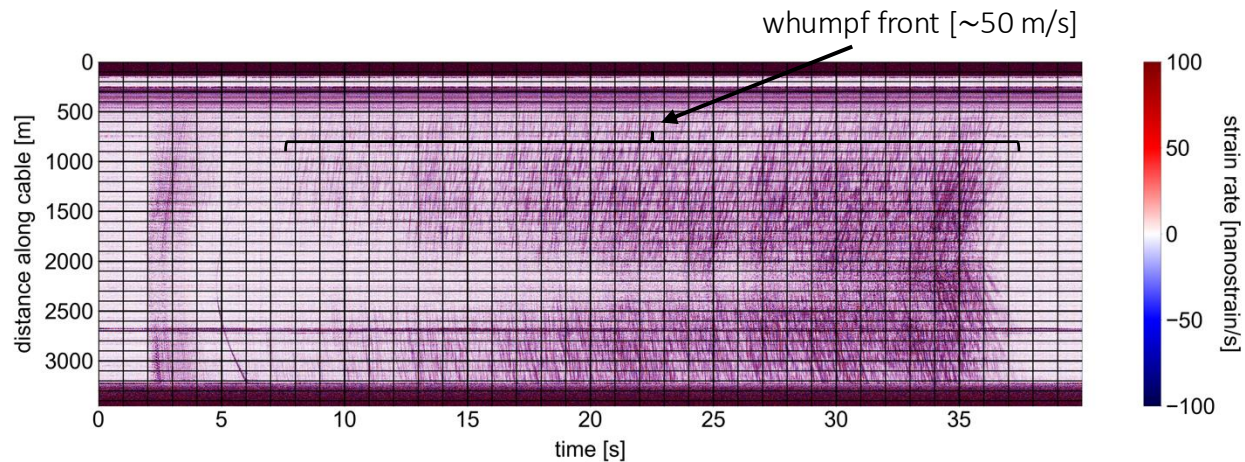


Jamieson & Schweizer, 2000

Slab cracking atop depth hoar layer



Knoff, 2016



Depth hoar layer



Jamieson & Schweizer, 2000

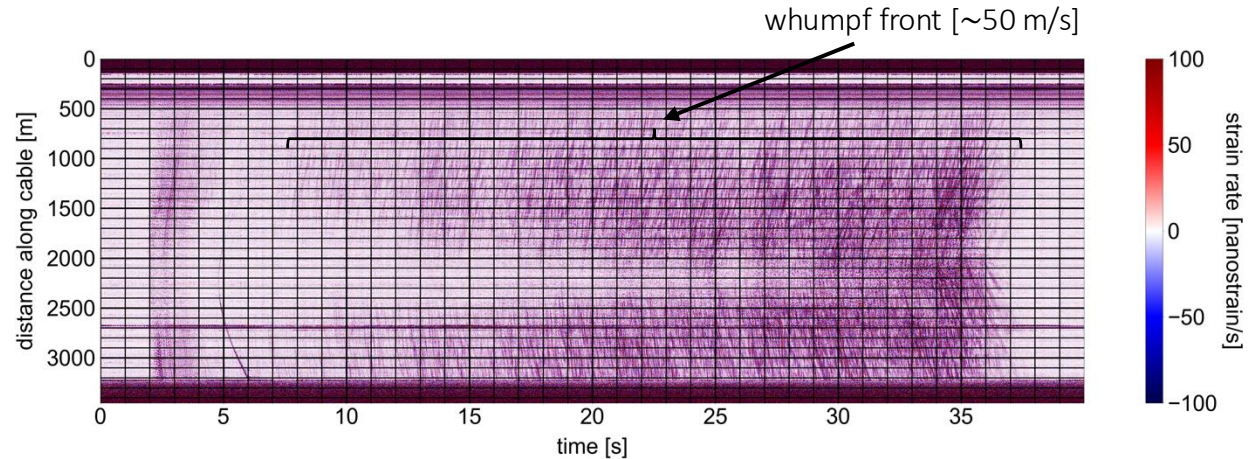
Slab cracking atop depth hoar layer



Knoff, 2016

- Rheologic properties of depth hoar
- Rupture kinematic & dynamics
- Avalanche triggering and dynamics

Have a chat with Janos Benz!



Talk structure

Part I

A brief survey of cryospheric DAS signals
and what we may learn from them about the underlying processes

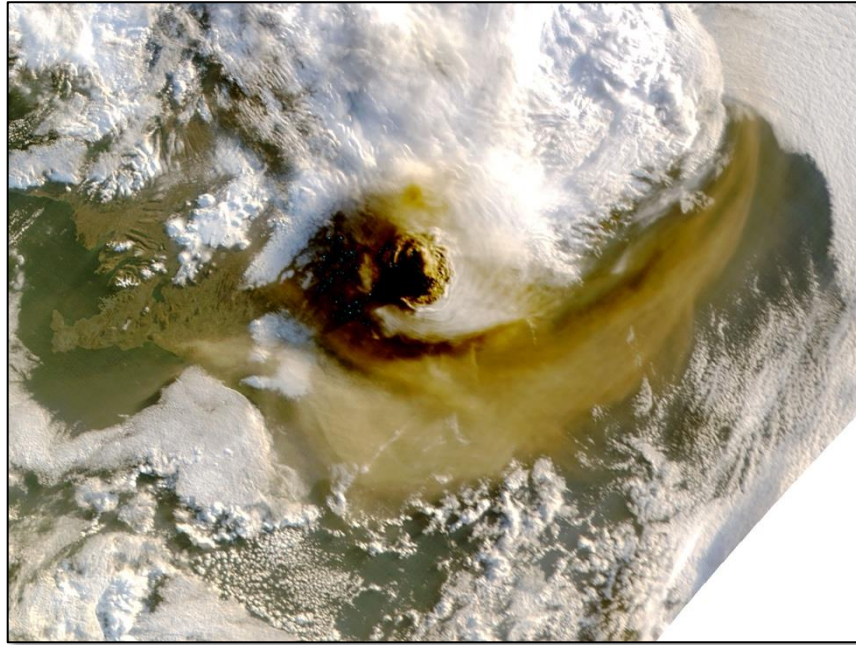
Part II

A closer look

- 1: Floating ice sheet resonance
- 2: Wave guiding in firn layers
- 3: Seismic cascades in ice streams

Floating ice sheet resonance: Grimsvötn, Iceland

in collaboration with Sara Klaasen, Kristín Jónsdóttir, Sölvi Thrastarson, Yeşim Çubuk-Sabuncu and Patrick Paitz



Satellite image of the ash plume during the 2011 eruption of Grimsvötn (NASA)

SETTING

- Iceland's most active volcano

Most recent major eruption in 2011

Caldera diameter around 10 km

- Subglacial

Located beneath Vatnajökull

Subglacial caldera lake

- Natural hazards

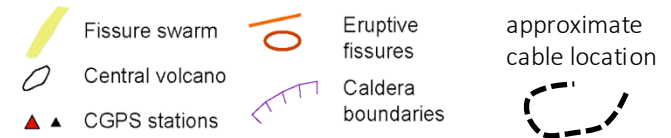
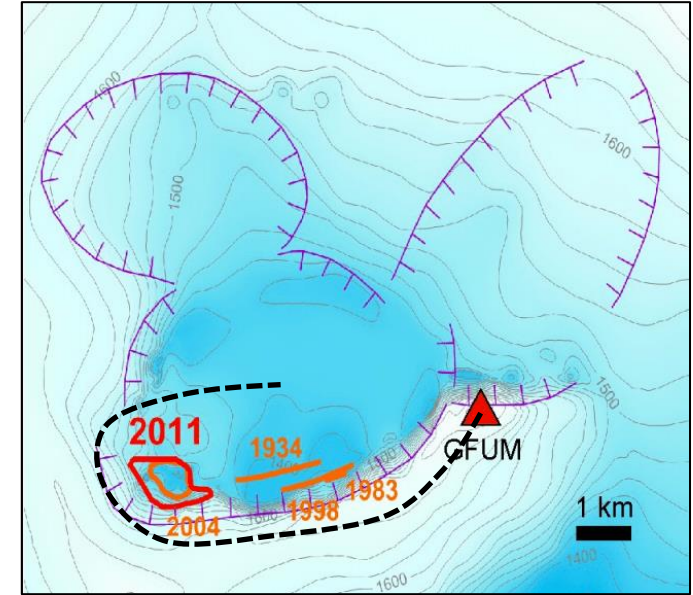
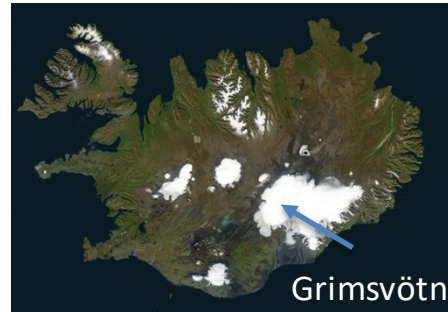
Subglacial floods (jökulhlaup)

Ash clouds and aviation

Climate impact

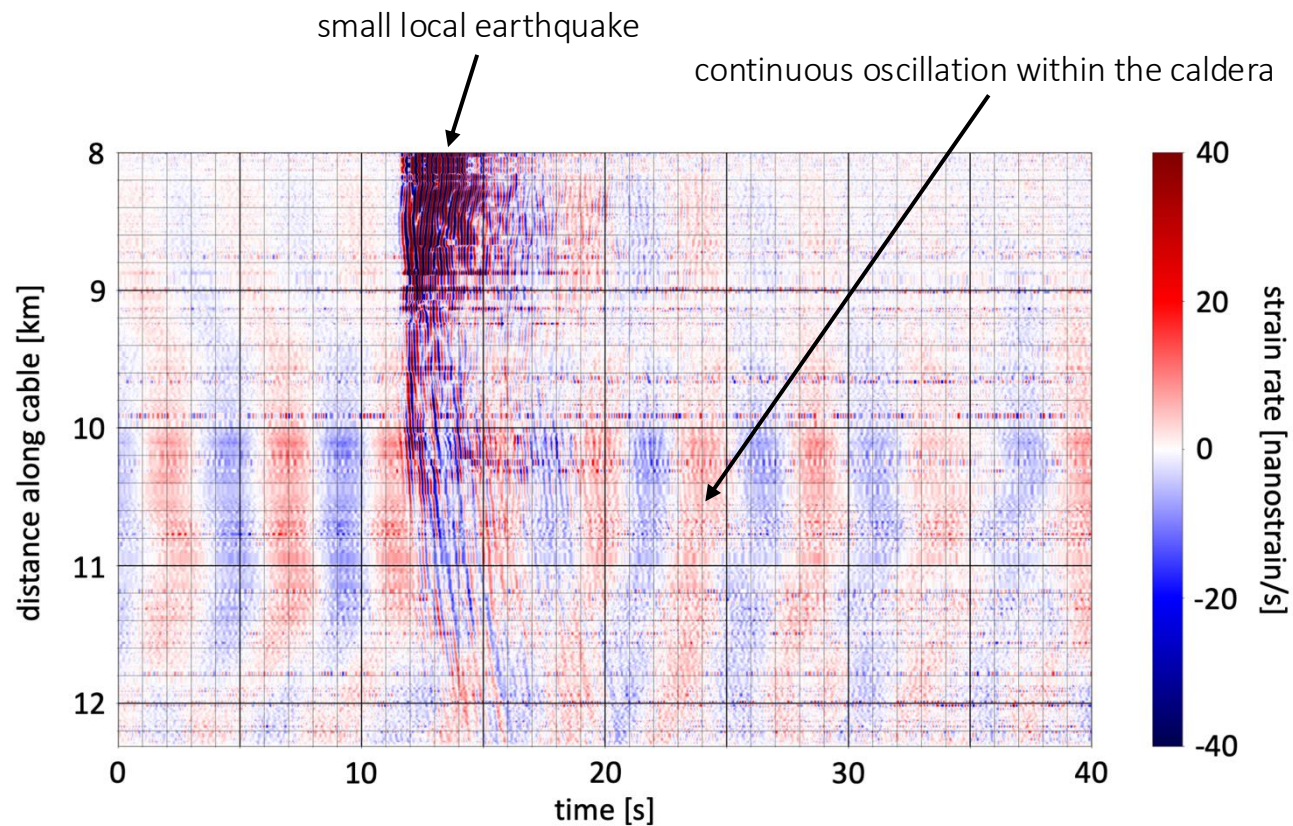
- Fiber-optic cable

>12 km long cable around and into the caldera



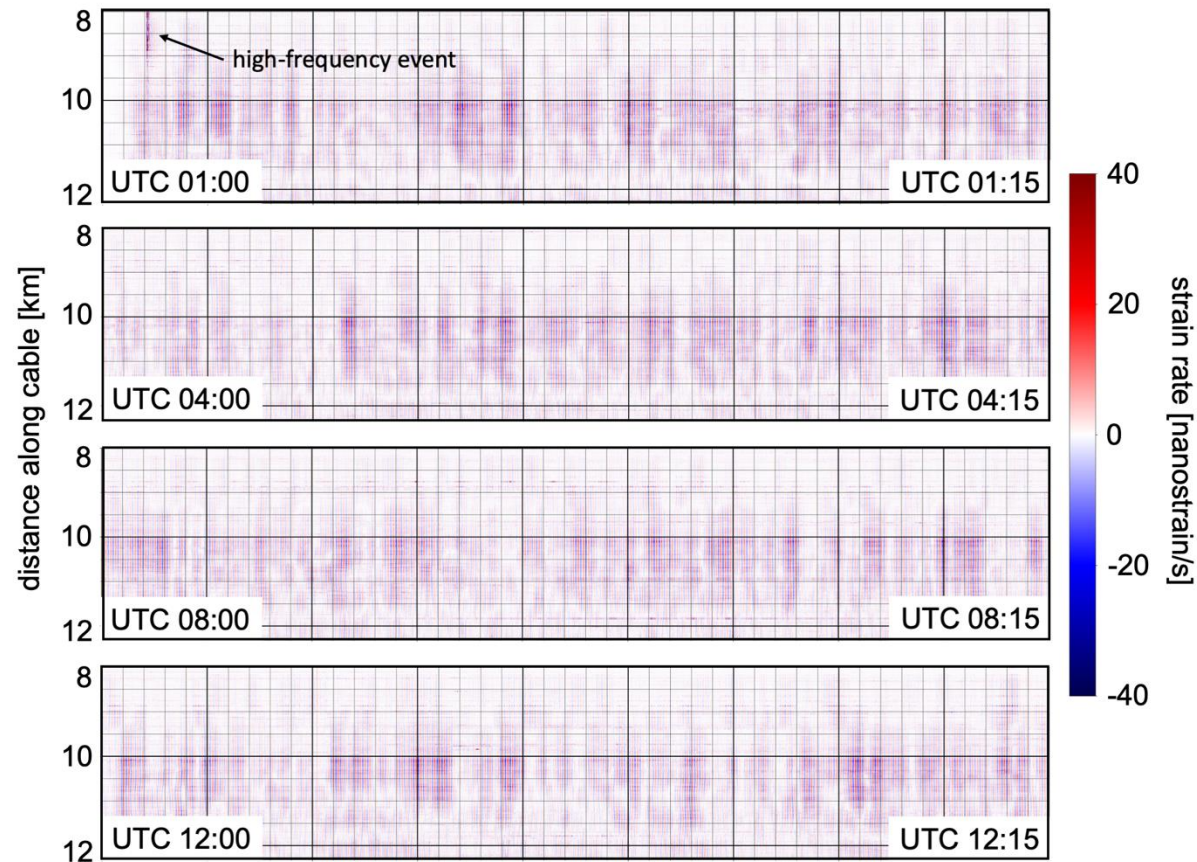
DATA EXAMPLE

40 s of data from 7 May 2021



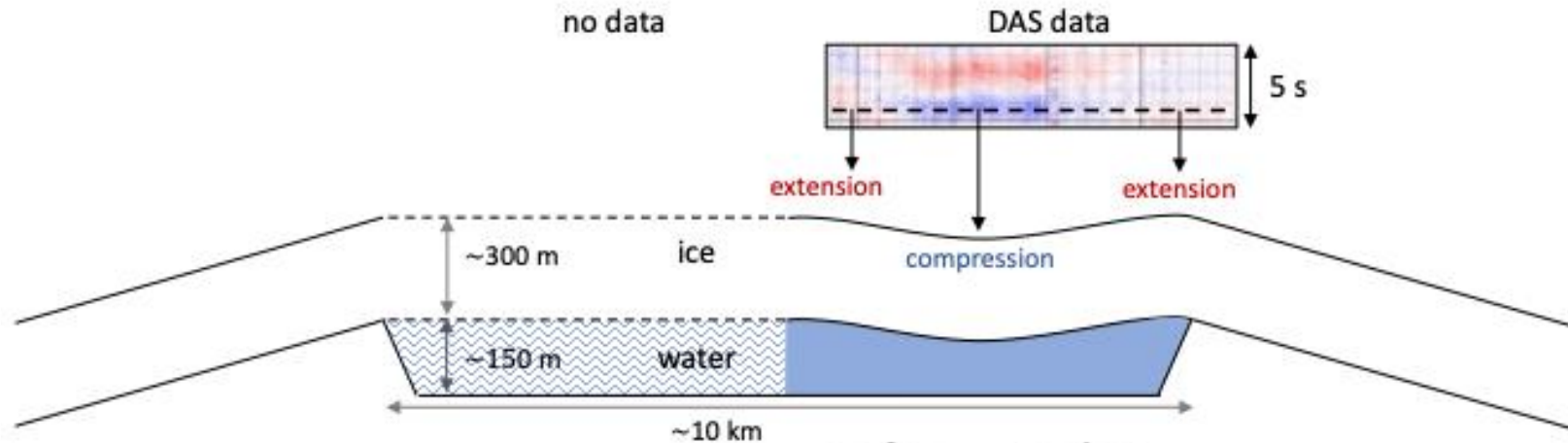
DATA EXAMPLE

12 h of data from 7 May 2021



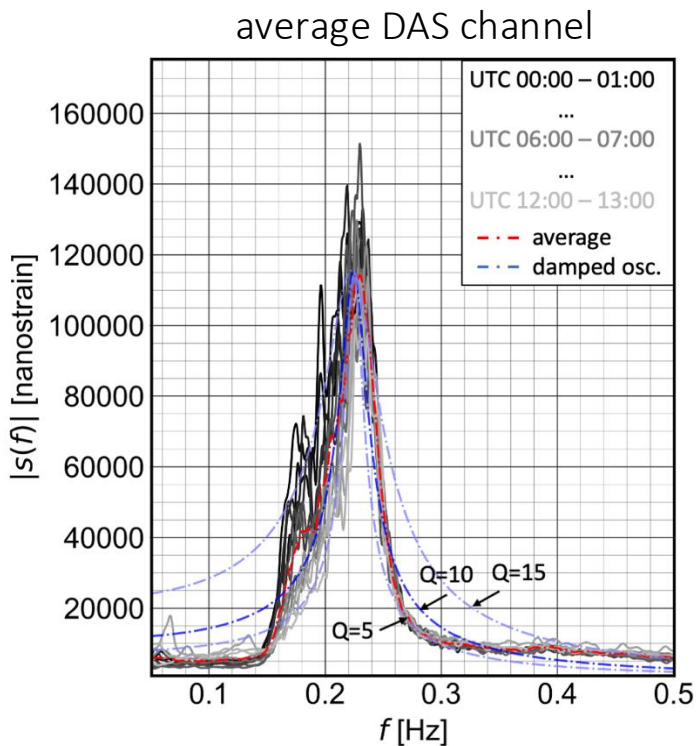
- Grimsvötn's caldera oscillates continuously.
- Oscillation frequency: ~ 0.2 Hz.
- What kind of wave could that be?

SCHEMATIC GEOMETRY



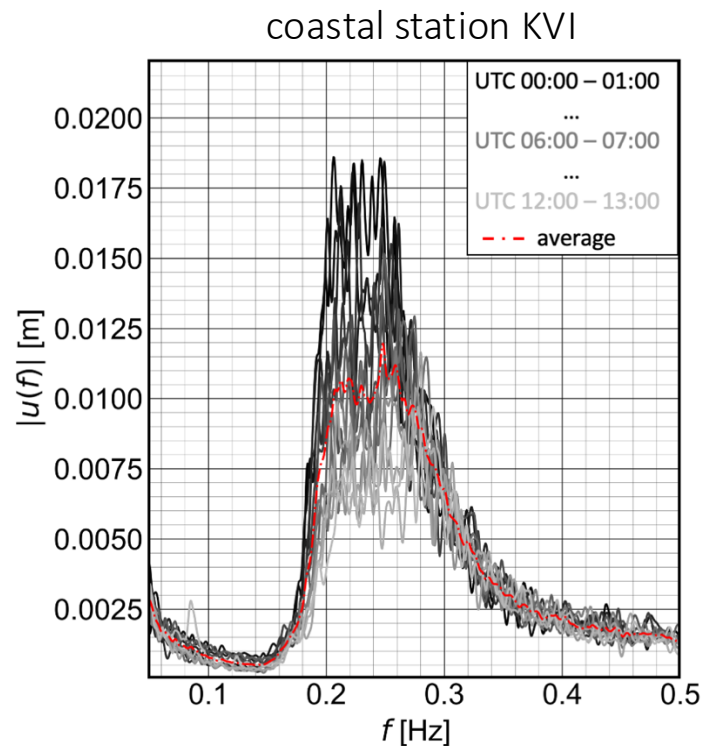
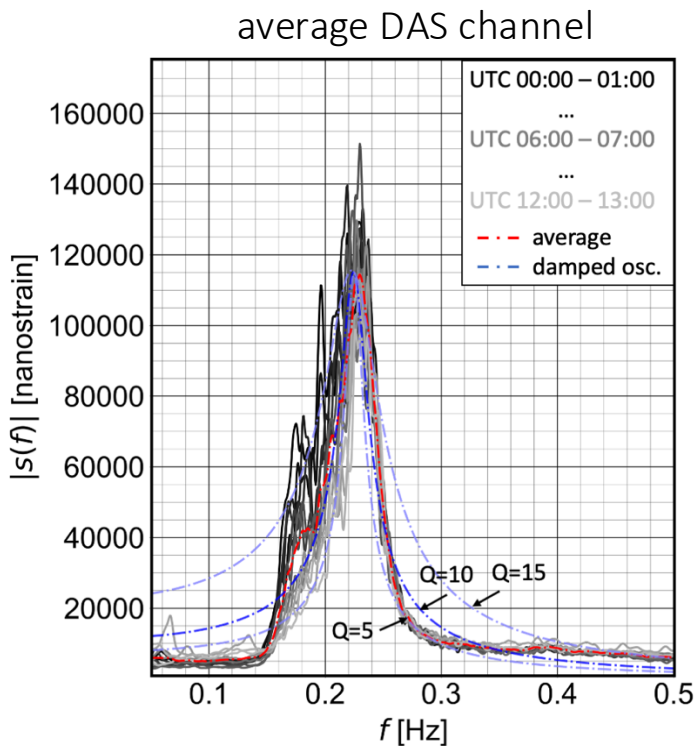
- Every ~ 5 s, the ice sheet goes through an up/down oscillation cycle.
- Could it be that ambient vibrations (caused by ocean waves) excite the ice sheet?

SPECTRA FROM 7 MAY 2021



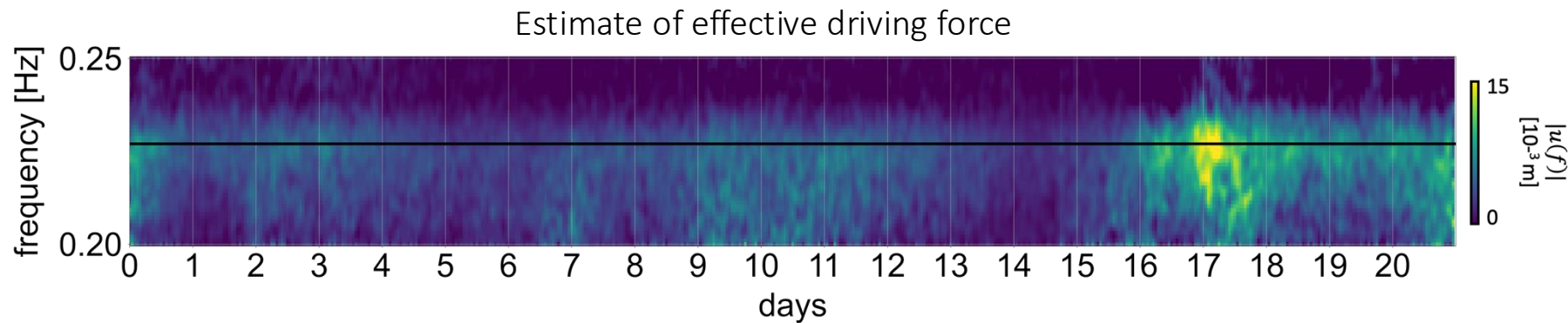
- Very narrow spectral peak
 - Spectrum nearly constant over time
 - Resembles spectrum of a damped harmonic oscillator with $Q \sim 10$
- Driving force required!

SPECTRA FROM 7 MAY 2021



- Spectral amplitude varies by a factor of ~ 3 within only 12 hours!
- Ambient vibrations unlikely to be the driving force

ESTIMATION OF DRIVING FORCES



- Low-frequency sources act almost continuously.
- May be related to volcanic tremor.

CONCLUSIONS

- Ice sheet resonance
 - amplification mechanism that makes otherwise invisible forces visible
 - large natural loudspeaker
- Attenuation requires nearly continuous driving forces
- May be due to volcanic tremor, e.g., related to geothermal activity.
- May offer new, and somewhat exotic, opportunities for volcano monitoring.

Fibre-optic airplane seismology: Northeast Greenland Icestream

in collaboration with Coen Hofstede and Brian L. N. Kennett

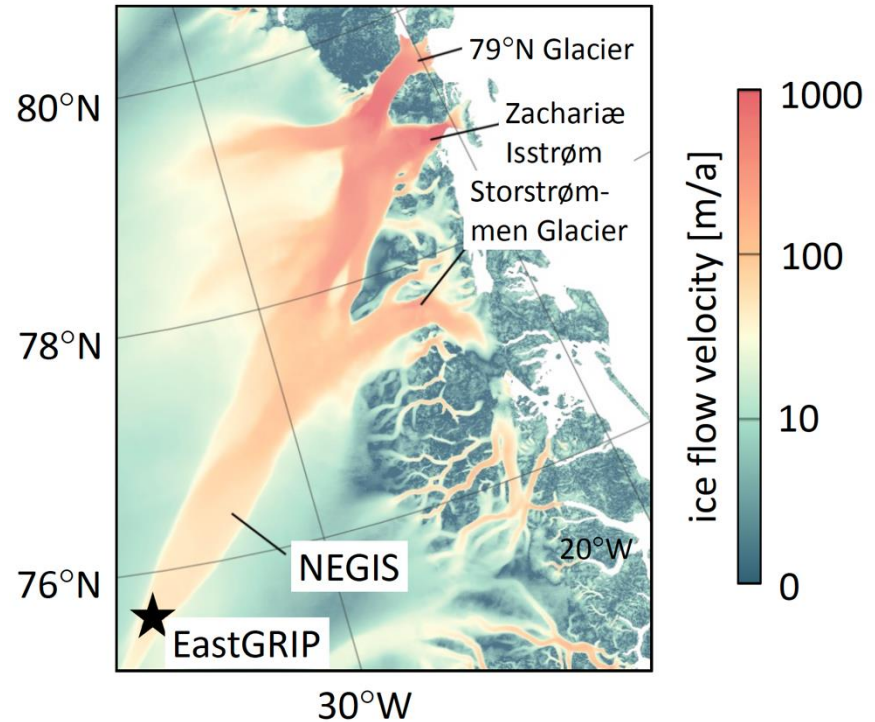


Midnight in July 2022 at the EastGRIP camp on the Northeast Greenland Icestream

SETTING

Northeast Greenland Ice Stream (NEGIS)

- Largest ice stream on the Greenland Ice Sheet, ~600 km
- Discharging ~12 % of ice sheet mass into North Atlantic
- Contribution to accelerated mass loss of the ice sheet



Surface velocities in NE Greenland derived from satellite data
[Joughin et al., 2016, 2018]

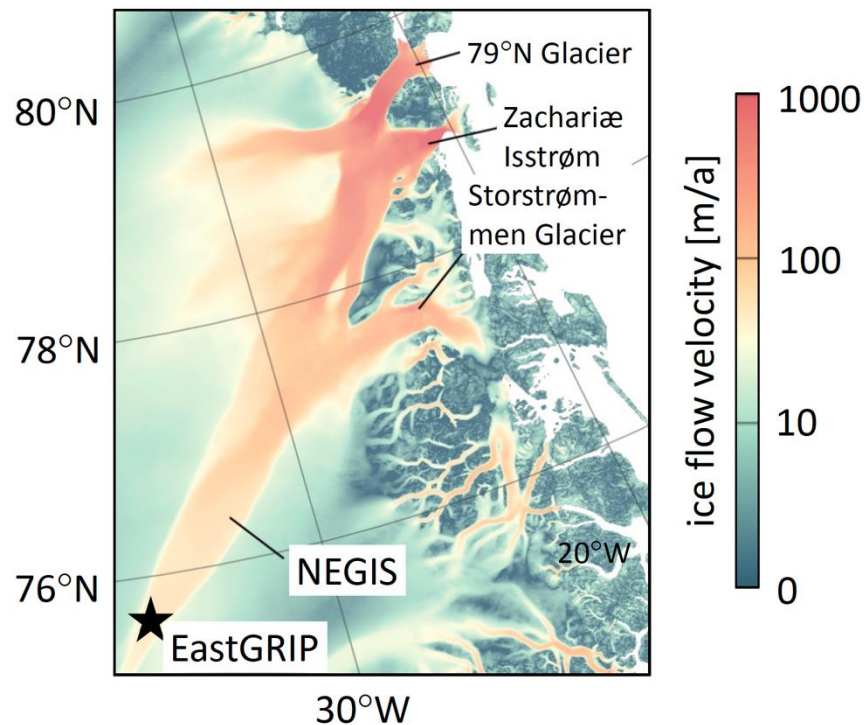
SETTING

Northeast Greenland Ice Stream (NEGIS)

- Largest ice stream on the Greenland Ice Sheet, ~600 km
- Discharging ~12 % of ice sheet mass into North Atlantic
- Contribution to accelerated mass loss of the ice sheet

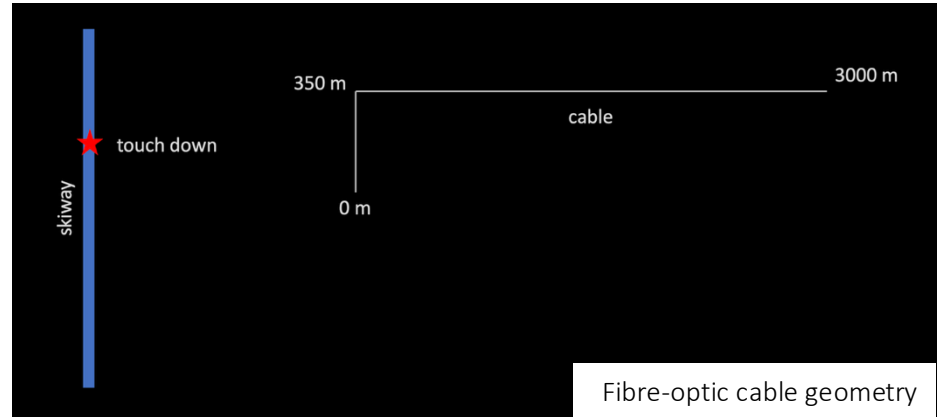
East Greenland Ice Core Project (EastGRIP)

- First retrieval of deep ice core from an active ice stream
- Ice thickness: ~2660 m
- Flow velocity: ~55 m / year
- Climate history and deformation patterns at depth

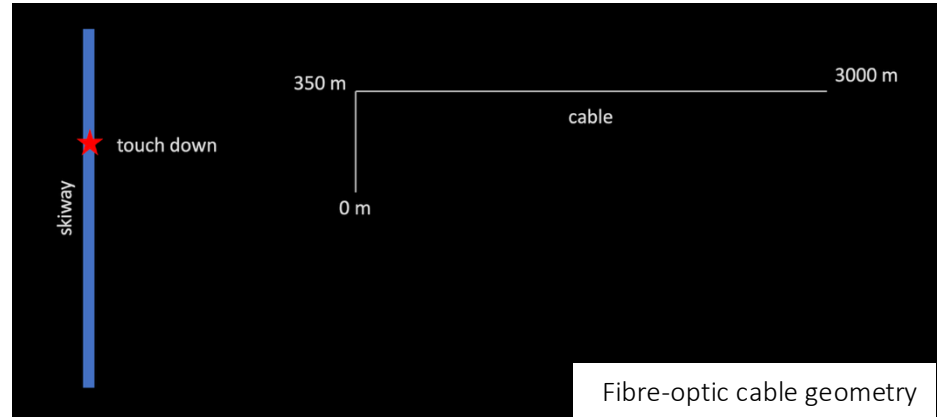


Surface velocities in NE Greenland derived from satellite data
[Joughin et al., 2016, 2018]

AIRPLANES AS SEISMIC SOURCES



AIRPLANES AS SEISMIC SOURCES



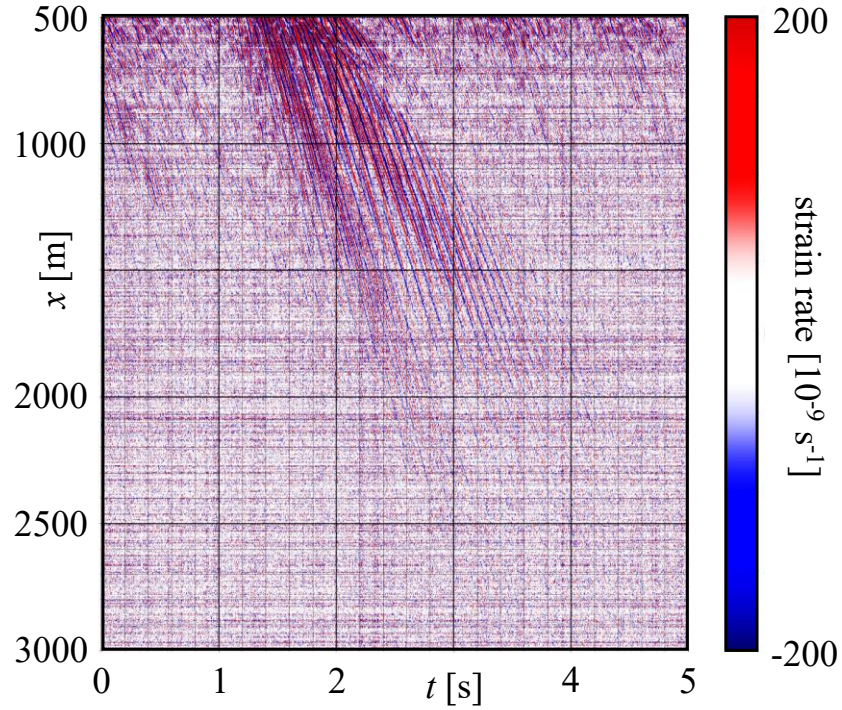
Technical details

- Sinking speed at landing ~ 1 m/s
- Estimated weight: $\sim 60\,000$ kg
- Kinetic energy: ~ 30 kJ
- Earthquake magnitude: ~ -0.2

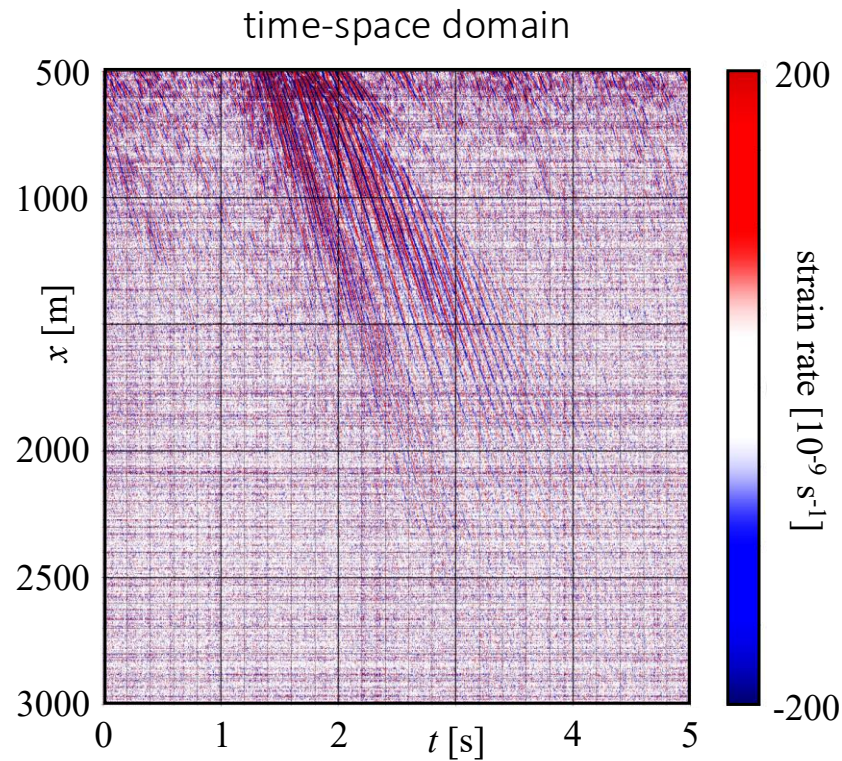


PHENOMENOLOGY

time-space domain



PHENOMENOLOGY

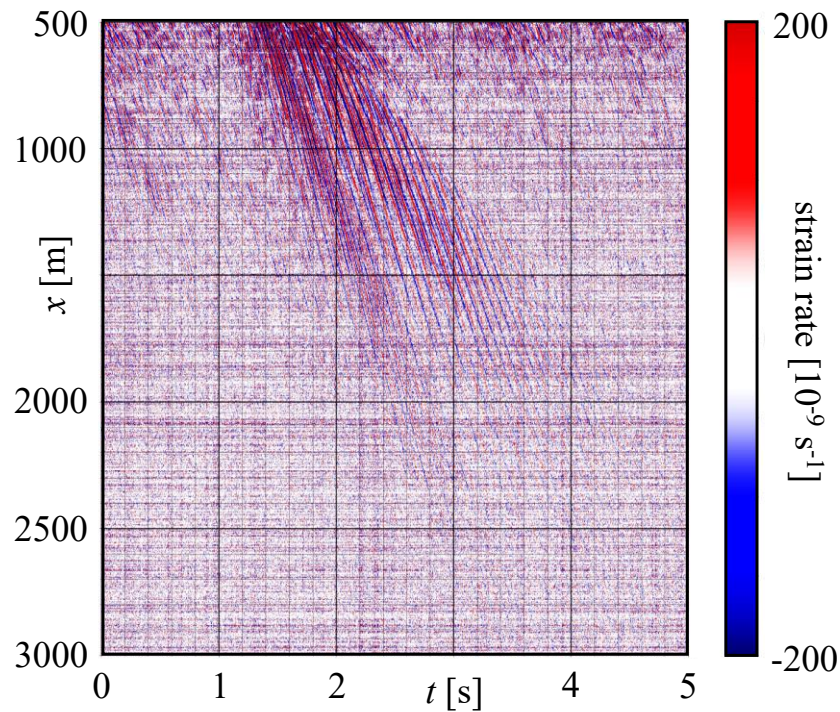


Dense spatial sampling →

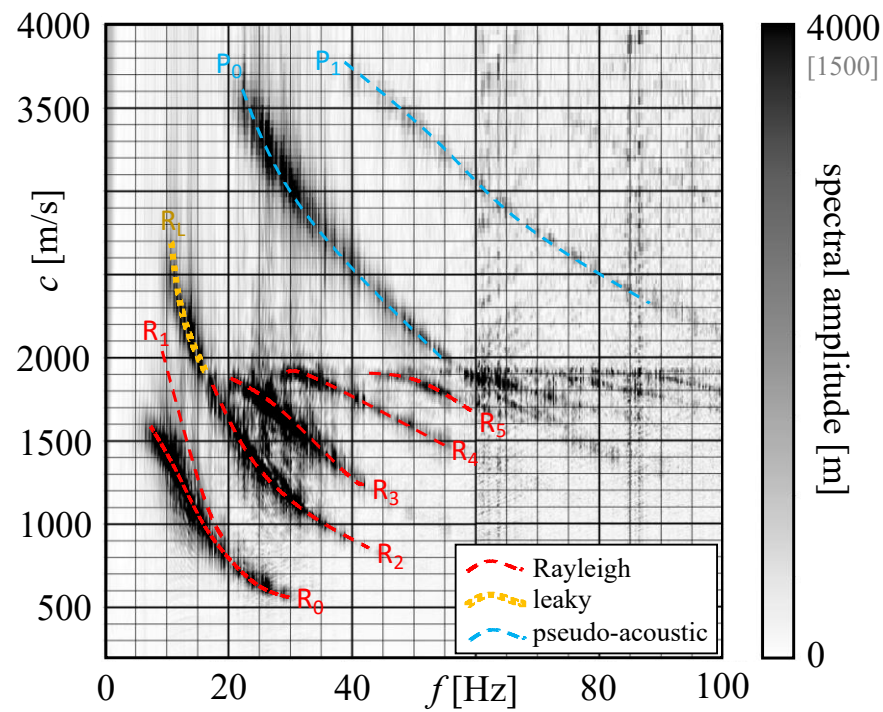
nearly perfect transformation into
velocity – frequency space.

PHENOMENOLOGY

time-space domain

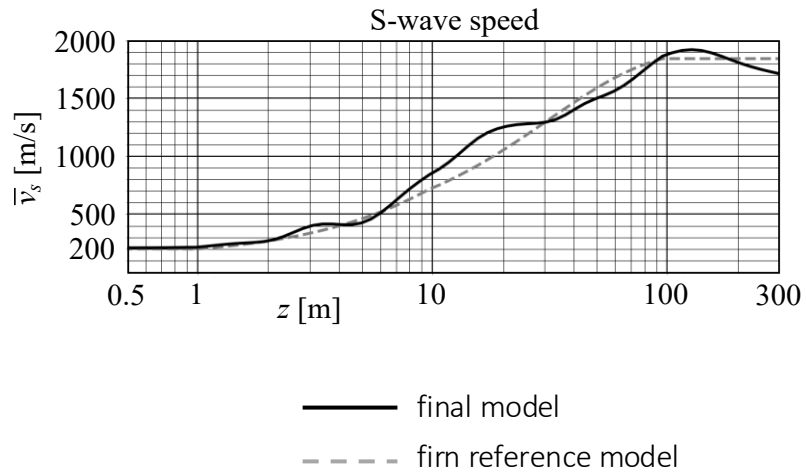


frequency-phase velocity domain



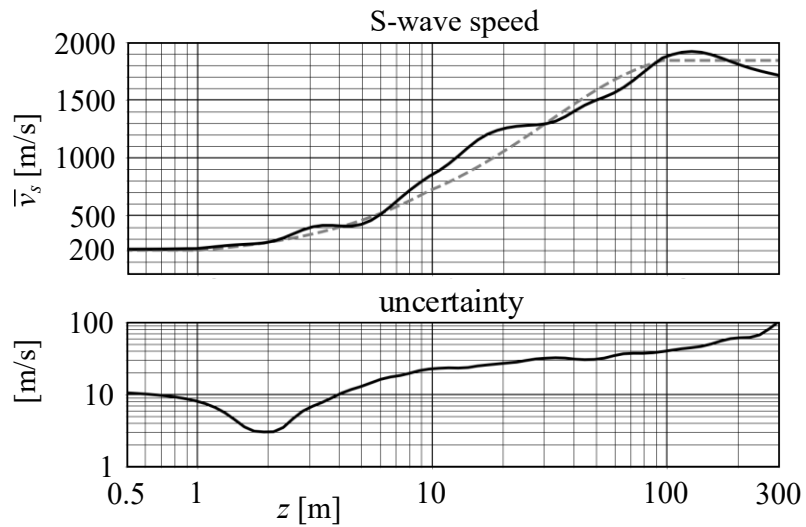
~15 visually detectable modes

BACKUS-GILBERT INVERSION



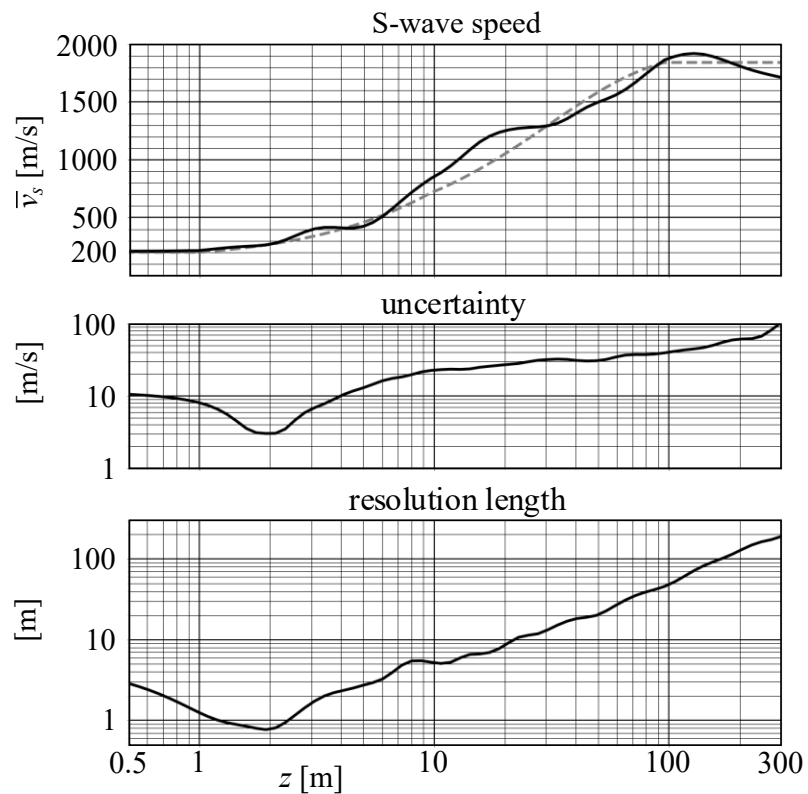
- S-wave speed model of the upper ~100 m

BACKUS-GILBERT INVERSION



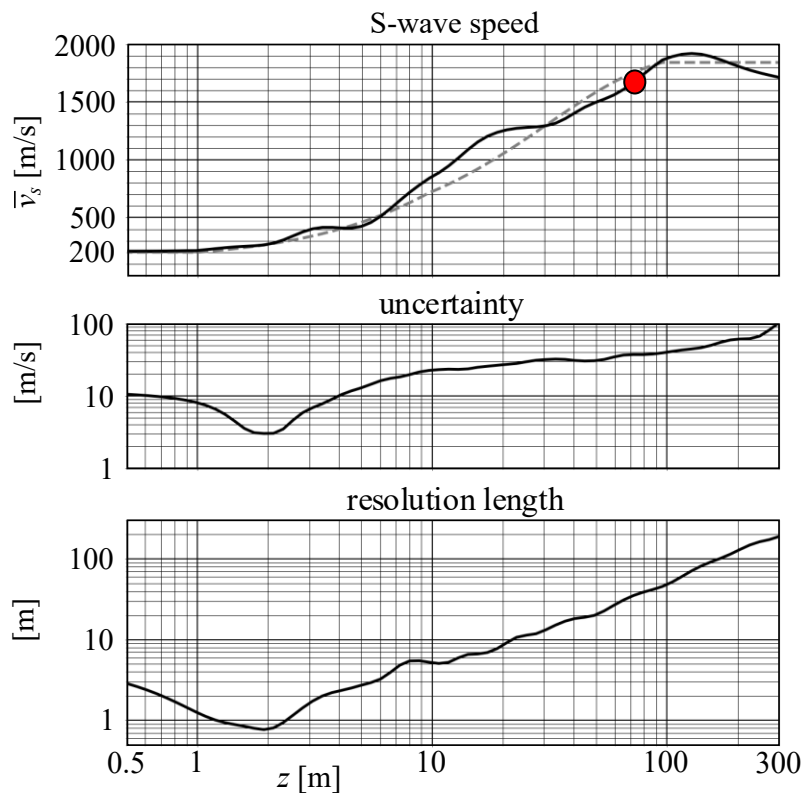
- S-wave speed model of the upper ~ 100 m
- Uncertainties of few to few tens of m/s.

BACKUS-GILBERT INVERSION



- S-wave speed model of the upper ~ 100 m
- Uncertainties of few to few tens of m/s.
- Resolution lengths of <1 to few tens of m.

BACKUS-GILBERT INVERSION



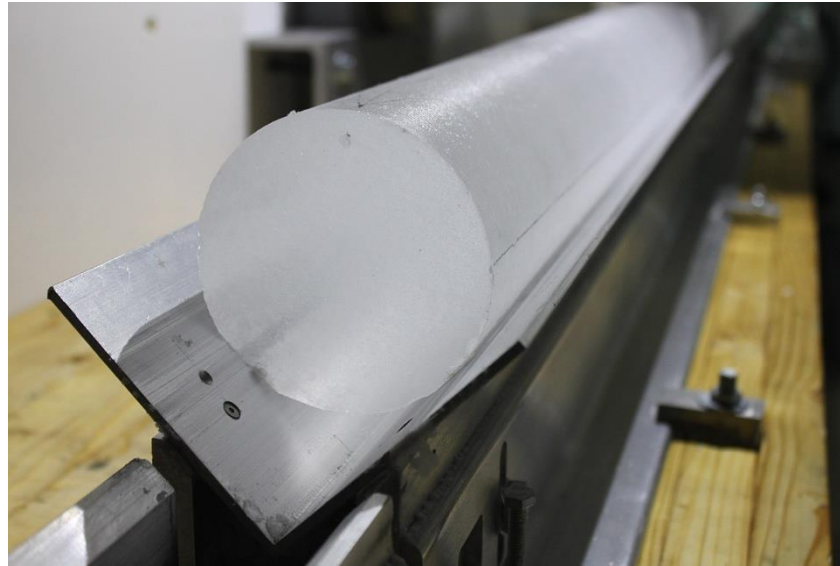
- S-wave speed model of the upper ~ 100 m
- Uncertainties of few to few tens of m/s.
- Resolution lengths of <1 to few tens of m.
- Estimate of firn-ice transition ● at 65 – 71 m depth [$v_s=1680$ m/s, 65 m from ice cores].

CONCLUSIONS

- The firn layer is an extremely good wave guide.
 - Exponential wave speed profile
 - Low attenuation
 - DAS + unconventional seismic source → very unusual data quality.
 - Around 15 easily detectable modes, many of which can be identified.
 - Large bandwidth from 3 – 100 Hz.
 - Backus-Gilbert inversion → very detailed firn model
 - Resolution lengths: few metres
 - Uncertainties: ~ 10 m/s
 - Unconventional contribution to
 - Ice sheet mass balance estimates
 - Ice core climatology
- Can be extended to direct imaging of glacial density. [Have a chat with Ariane Lanteri]

Hidden Cascades of Seismic Ice Stream Deformation

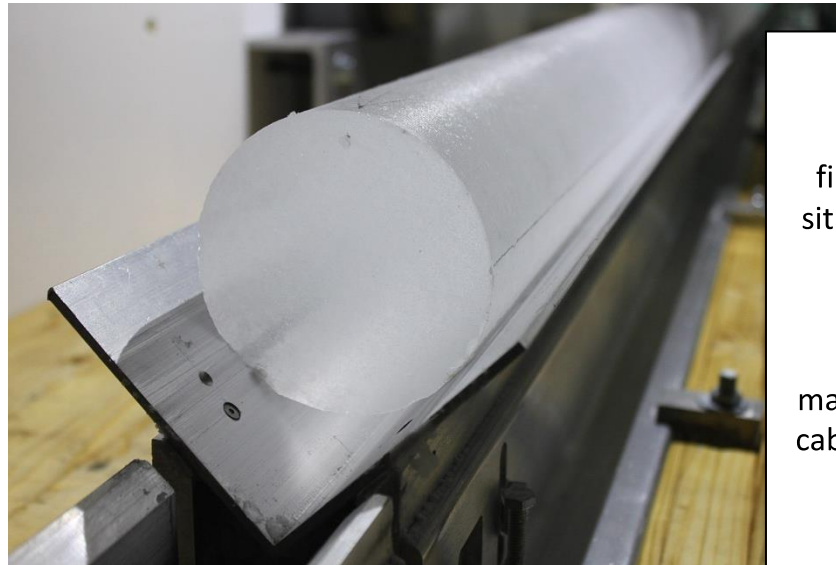
Andreas Fichtner, Coen Hofstede, Brian L. N. Kennett, Anders Svensson, Julien Westhoff, Fabian Walter, Jean-Paul Ampuero, Eliza Cook, Dimitri Zigone, Daniela Jansen & Olaf Eisen



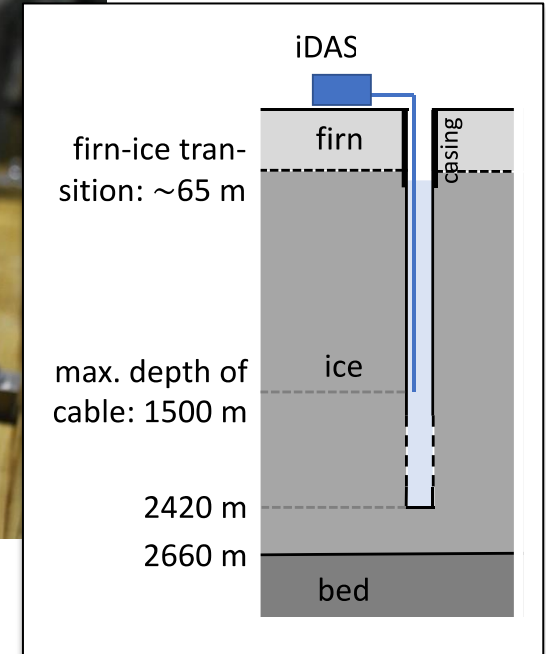
Fresh ice core retrieved from the EastGRIP borehole

Hidden Cascades of Seismic Ice Stream Deformation

Andreas Fichtner, Coen Hofstede, Brian L. N. Kennett, Anders Svensson, Julien Westhoff, Fabian Walter, Jean-Paul Ampuero, Eliza Cook, Dimitri Zigone, Daniela Jansen & Olaf Eisen

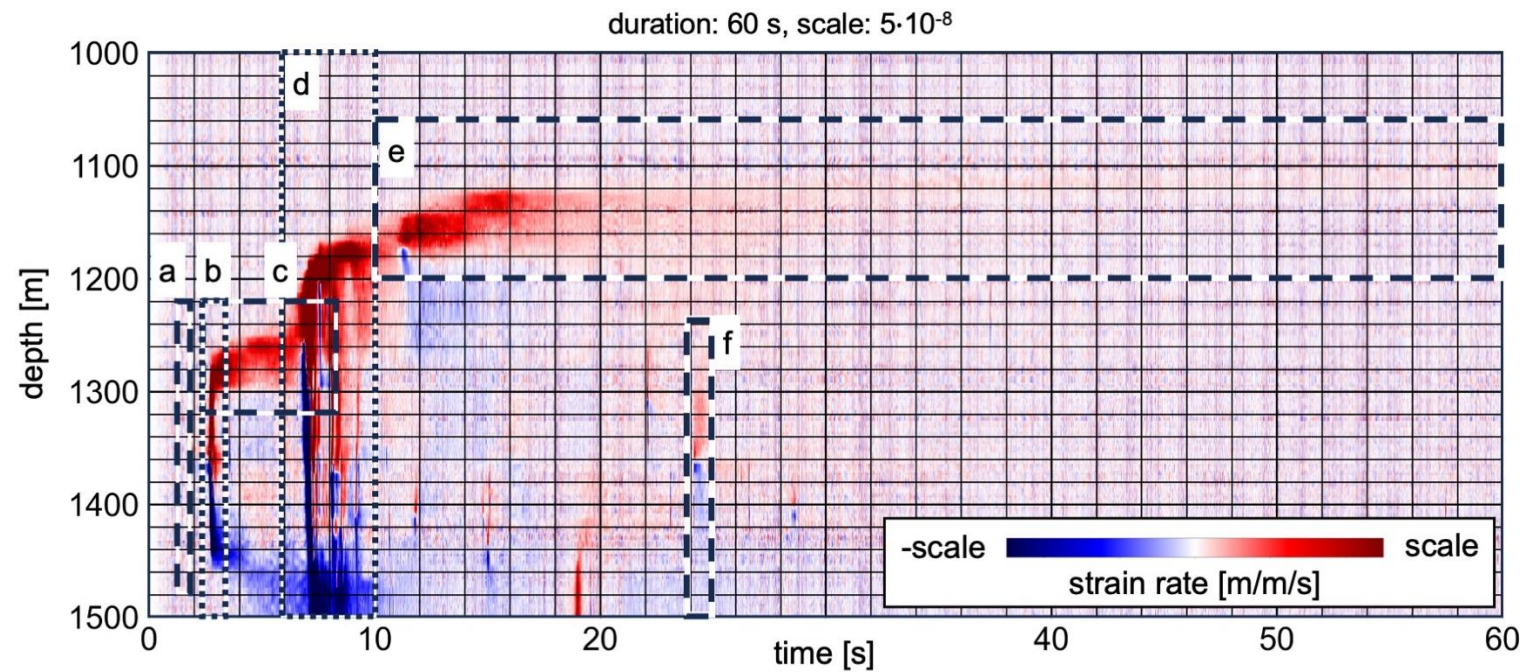


Fresh ice core retrieved from the EastGRIP borehole

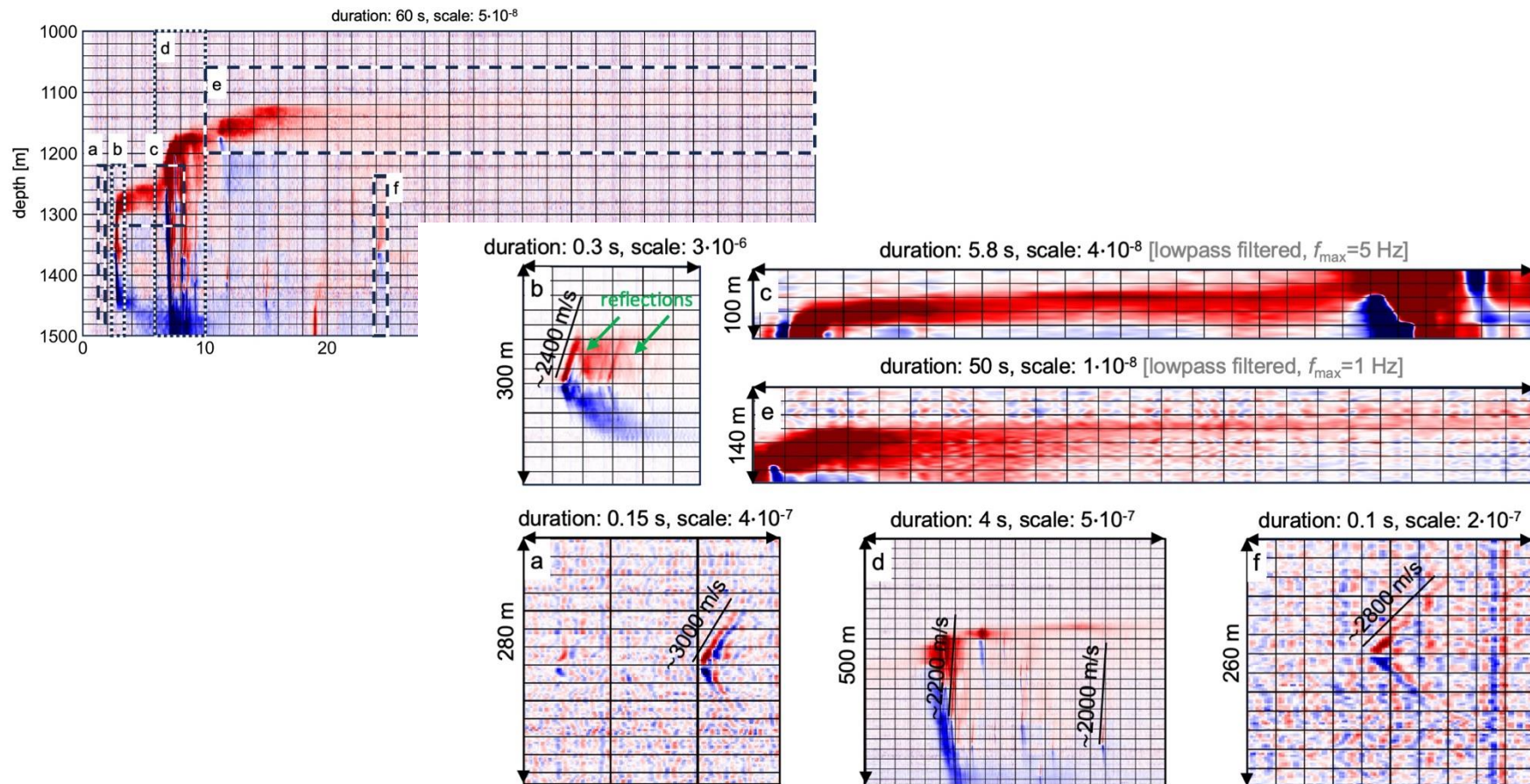


Phenomenology

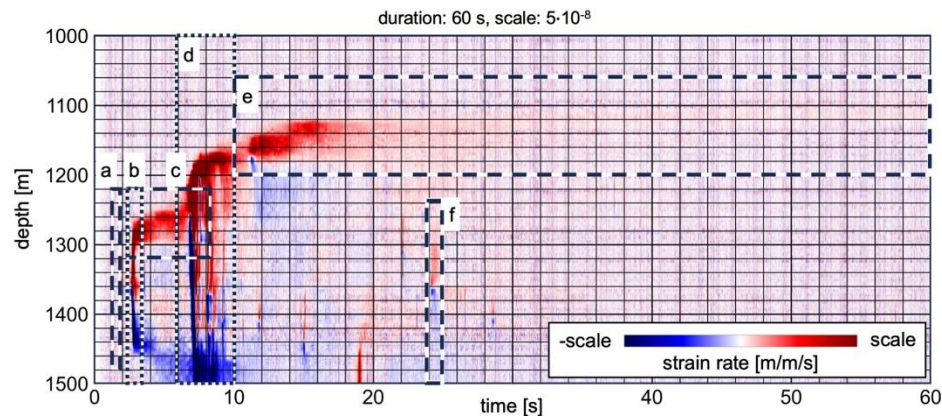
EVENT SEQUENCE 1 [STARTING UTC 2022.08.10 12:19:36]



EVENT SEQUENCE 1 [STARTING UTC 2022.08.10 12:19:36]

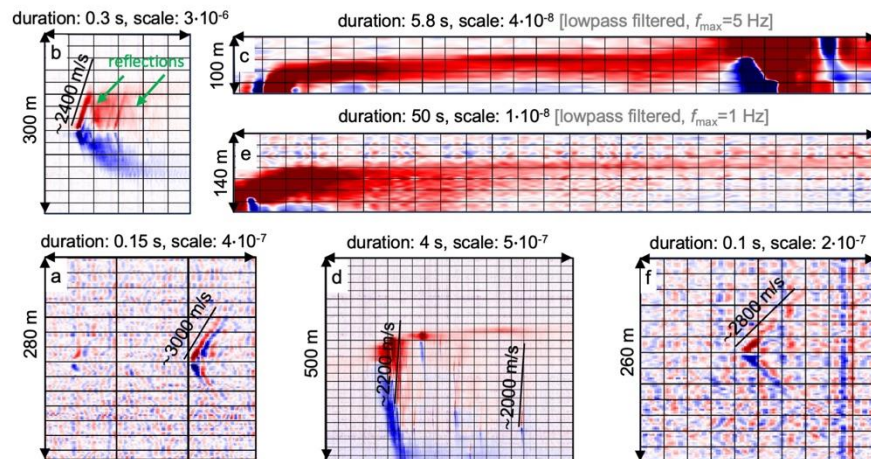


EVENT SEQUENCE 1 [STARTING UTC 2022.08.10 12:19:36]



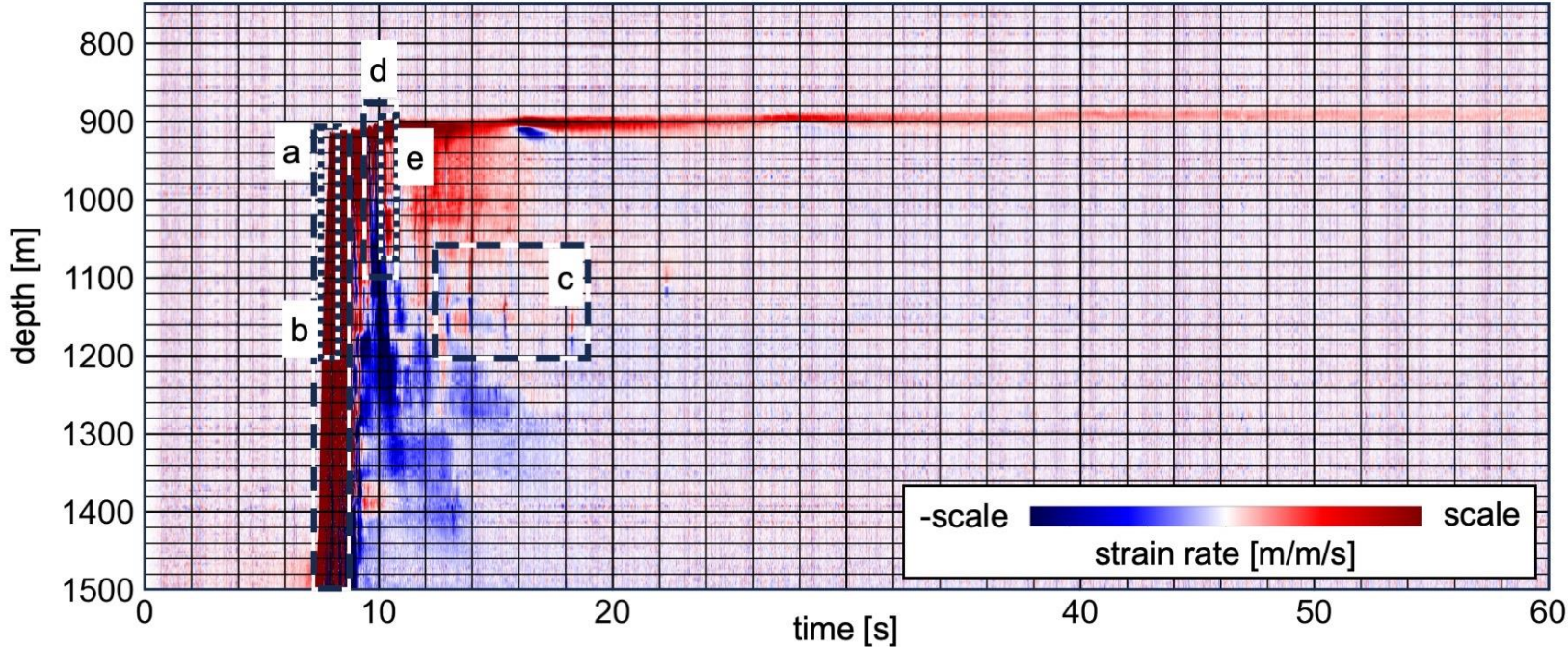
Selected elements of the wavefield

1. Sequences of tens of events.
2. Long-lasting [~ 1 min] localised strain accumulation.
3. Wave speeds: $\sim 1.8 - 3.0$ km/s [> 4 km/s for cable waves].

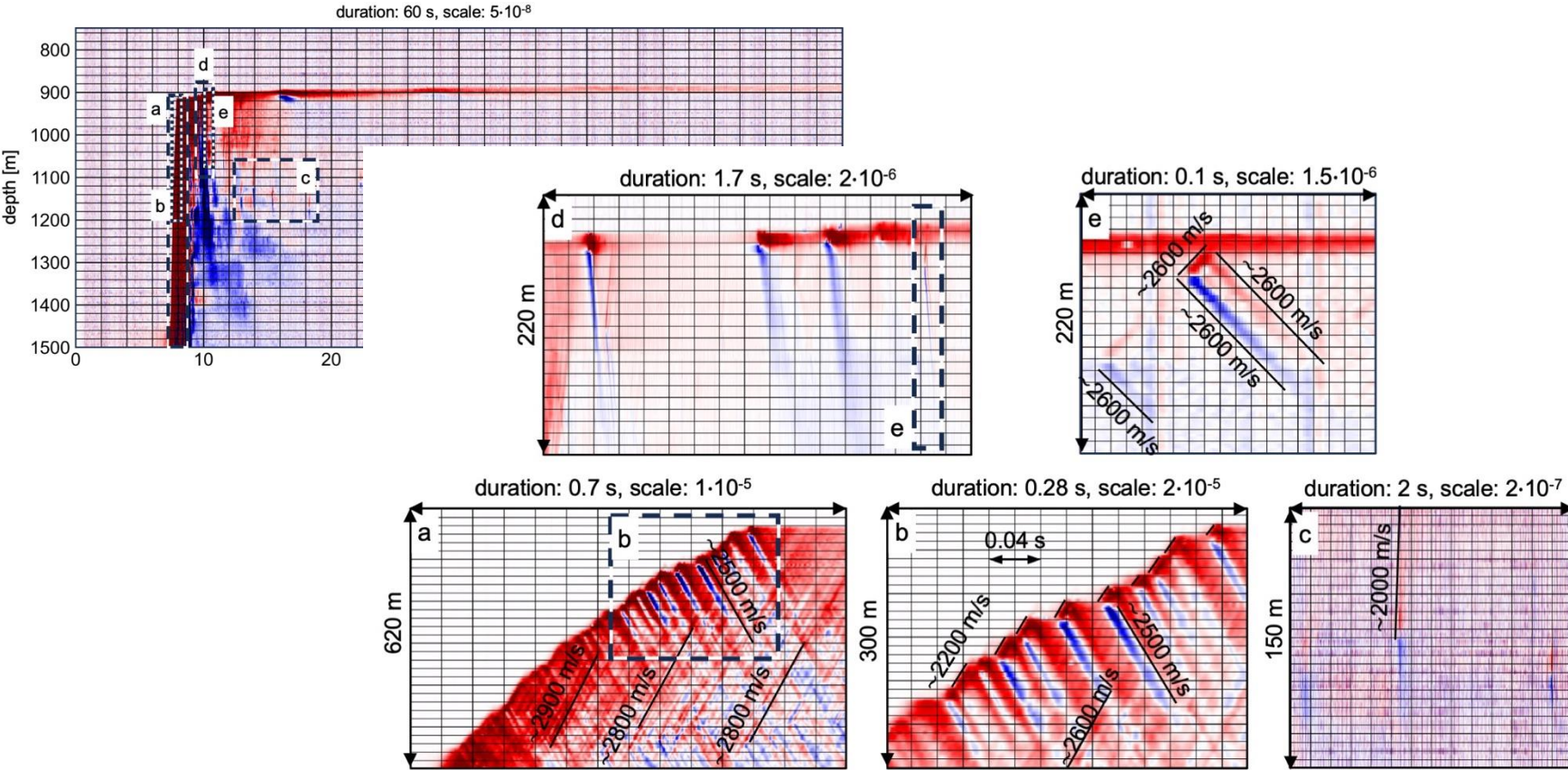


EVENT SEQUENCE 2 [STARTING UTC 2022.08.10 12:27:22]

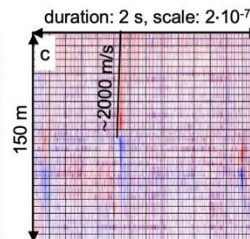
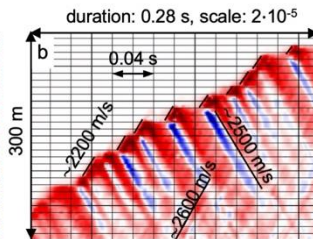
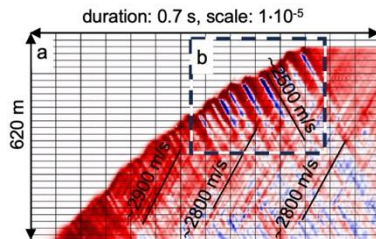
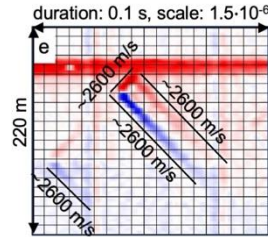
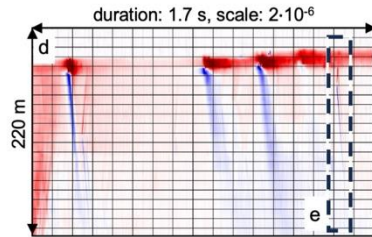
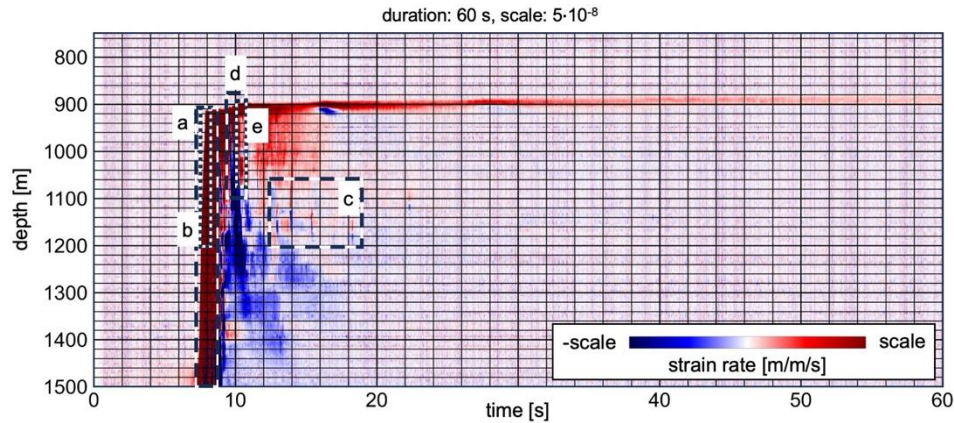
duration: 60 s, scale: $5 \cdot 10^{-8}$



EVENT SEQUENCE 2 [STARTING UTC 2022.08.10 12:27:22]



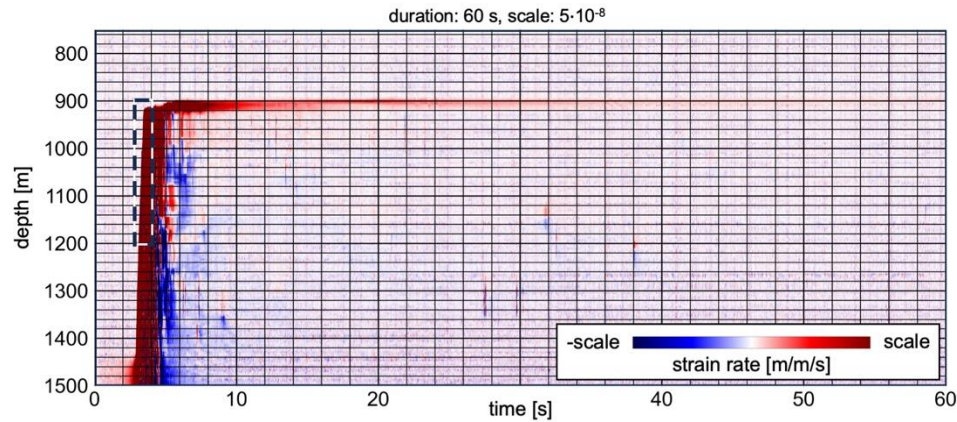
EVENT SEQUENCE 2 [STARTING UTC 2022.08.10 12:27:22]



Selected elements of the wavefield

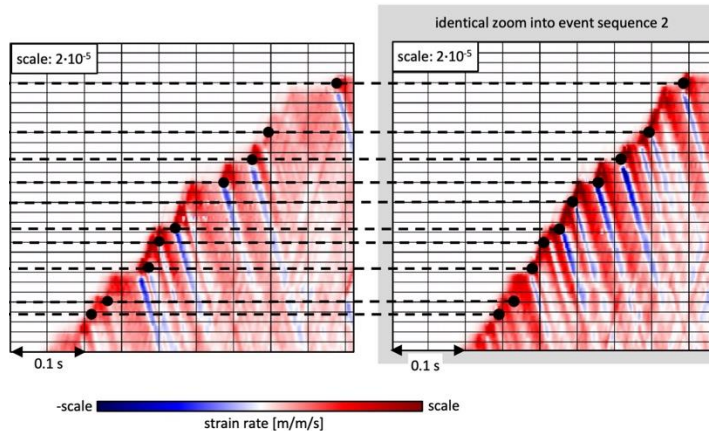
1. Sequences of tens of events.
2. Long-lasting [~ 1 min] localised strain accumulation.
3. Wave speeds: $\sim 1.8 - 3.0$ km/s [> 4 km/s for cable waves].
4. Event initiates below the downhole cable end.
5. Subevents trigger each other; ~ 100 of them.

EVENT SEQUENCE 5 [STARTING UTC 2022.08.10 17:36:42]

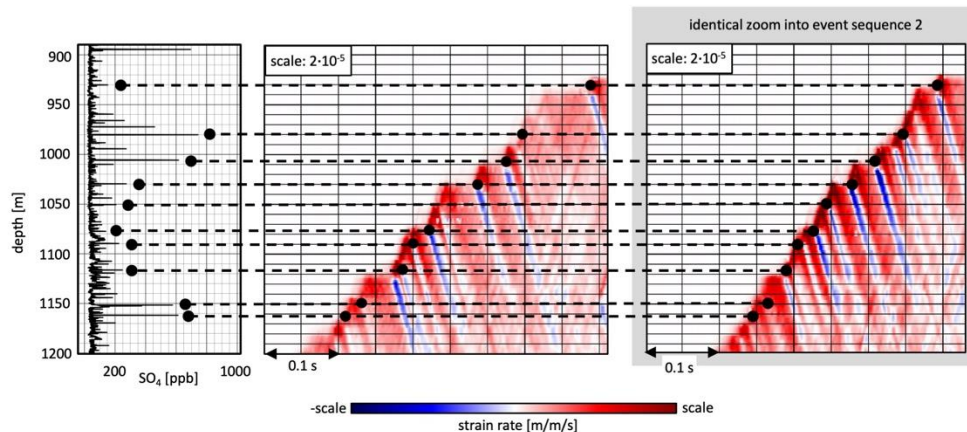
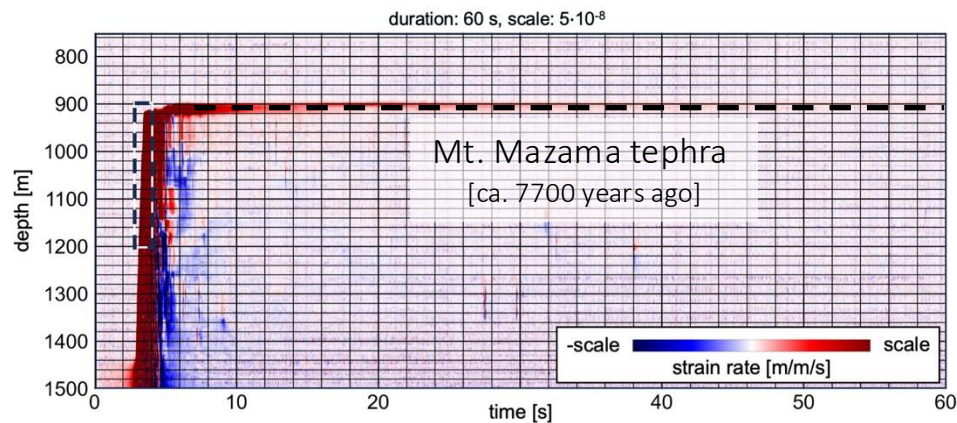


Selected elements of the wavefield

1. Sequences of tens of events.
2. Long-lasting [~ 1 min] localised strain accumulation.
3. Wave speeds: $\sim 1.8 - 3.0$ km/s [> 4 km/s for cable waves].
4. Event initiates below the downhole cable end.
5. Subevents trigger each other; ~ 100 of them.
6. Large complex events may repeat almost exactly.



EVENT SEQUENCE 5 [STARTING UTC 2022.08.10 17:36:42]



Selected elements of the wavefield

1. Sequences of tens of events.
2. Long-lasting [~ 1 min] localised strain accumulation.
3. Wave speeds: $\sim 1.8 - 3.0$ km/s [> 4 km/s for cable waves].
4. Event initiates below the downhole cable end.
5. Subevents trigger each other; ~ 100 of them.
6. Large complex events may repeat almost exactly.

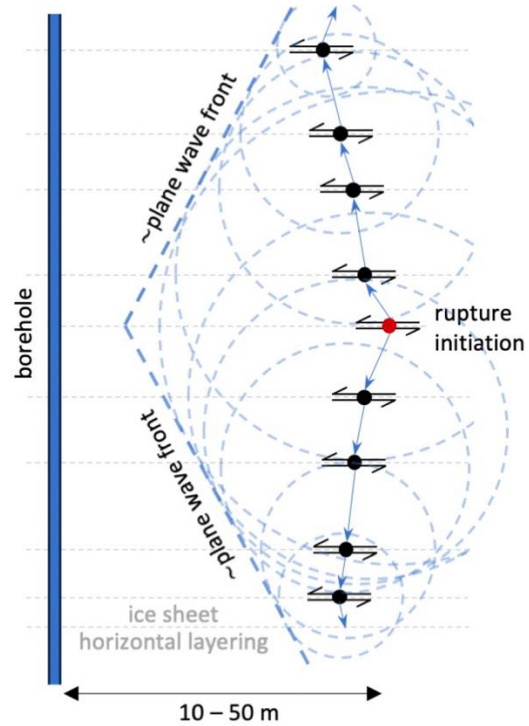
Correlation with volcanic impurities

1. Strain accumulation at Mt. Mazama tephra layer.
2. Smaller events preferentially initiate near volcanism-related SO₄ spikes in the ice.

Models & Links

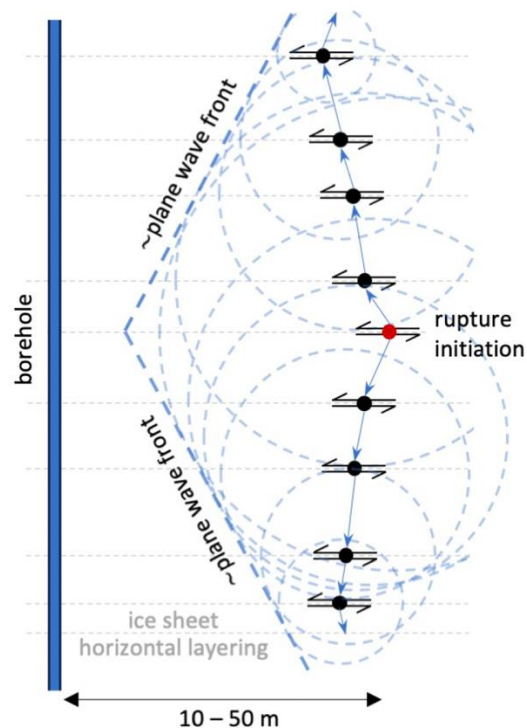
MODELLING VERTICAL EVENT CASCADES

Schematic rupture mechanism

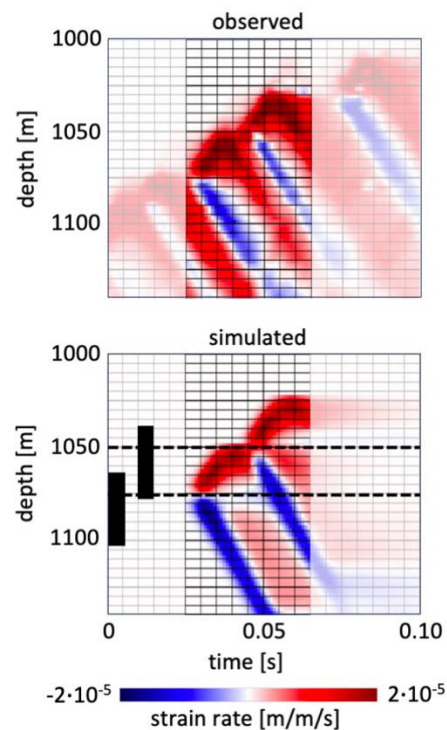


LINKS TO INDEPENDENT OBSERVATIONS

Schematic rupture mechanism

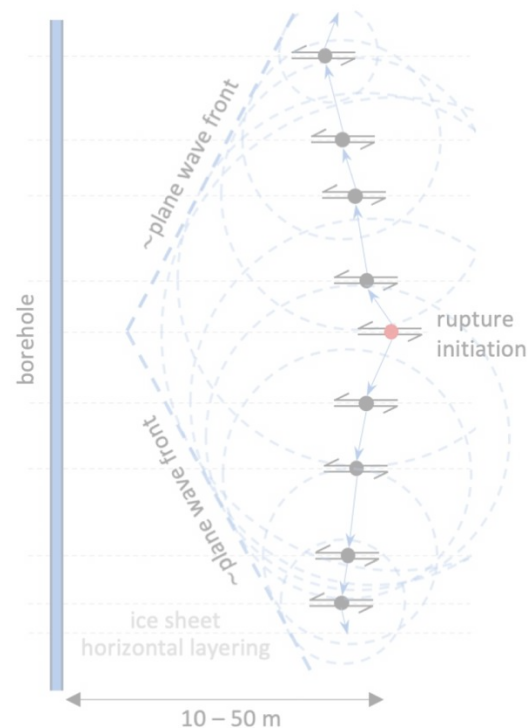


Waveform comparison

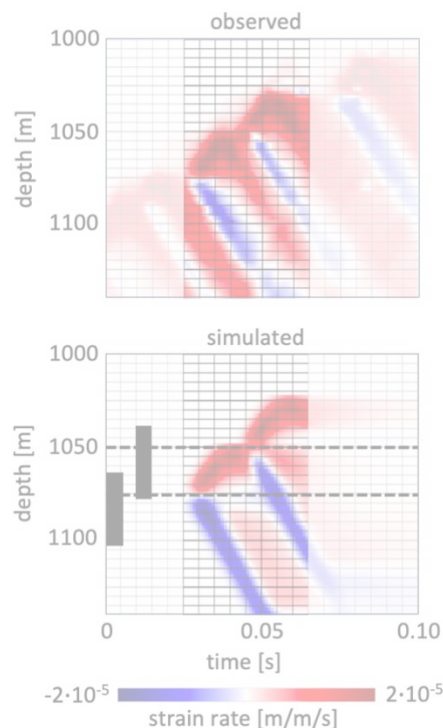


LINKS TO INDEPENDENT OBSERVATIONS

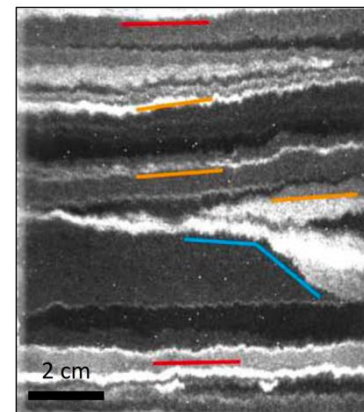
Schematic rupture mechanism



Waveform comparison



Ice deformation



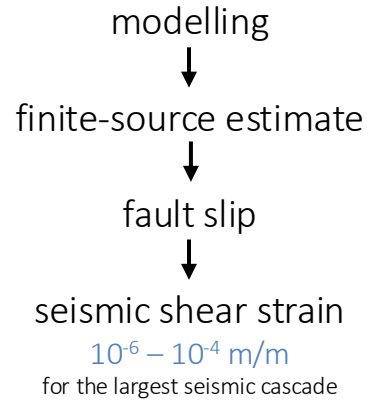
Brittle deformation structure
[thrust fault] from 1690 m depth

Lab experiments on SO_4 in ice:

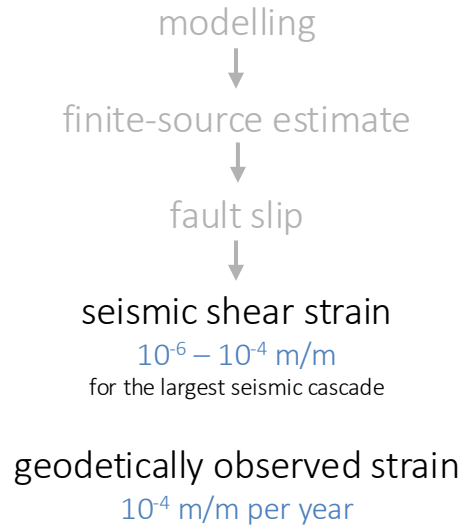
- reduction in viscosity
[Trickett et al., 2000]
- enhanced micro-cracking
[Hammonds & Baker, 2018]

Implications

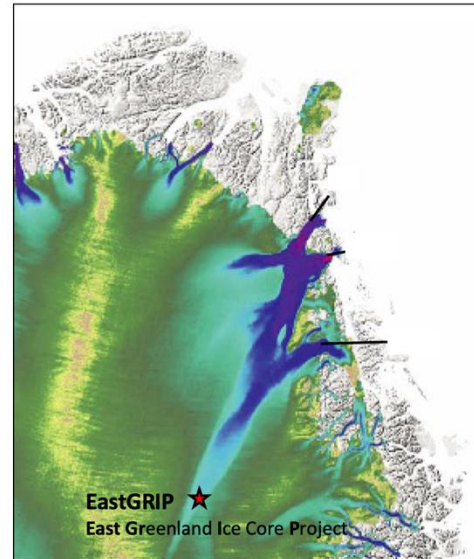
STRAIN ACCUMULATION



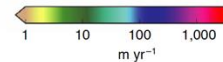
STRAIN ACCUMULATION



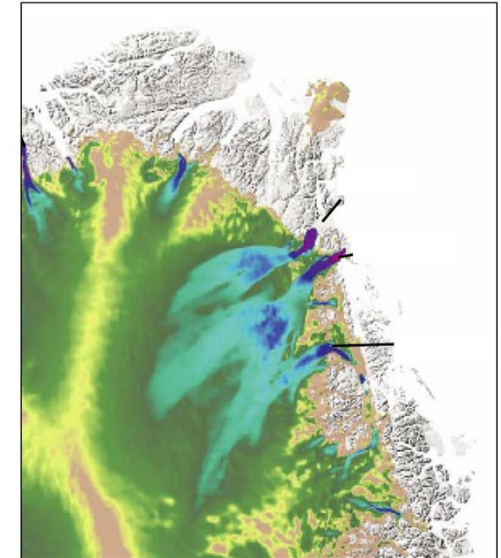
observed



Surface velocity [Aschwanden et al., 2016]



simulated
assuming viscous flow

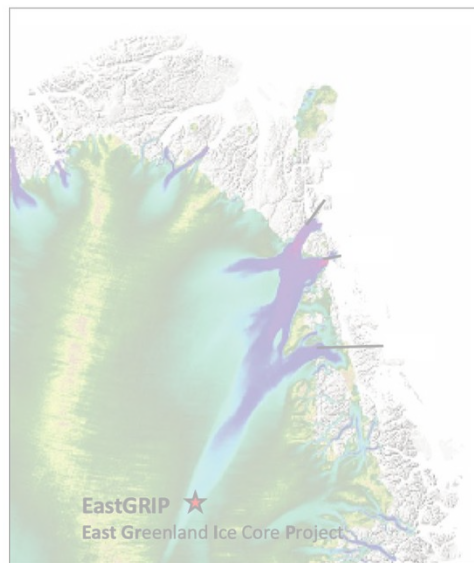


CONCLUSIONS

~100 event cascades per year
could fill the simulation-
observation gap!

Newly discovered brittle
deformation mode likely to be
relevant for ice stream
deformation.

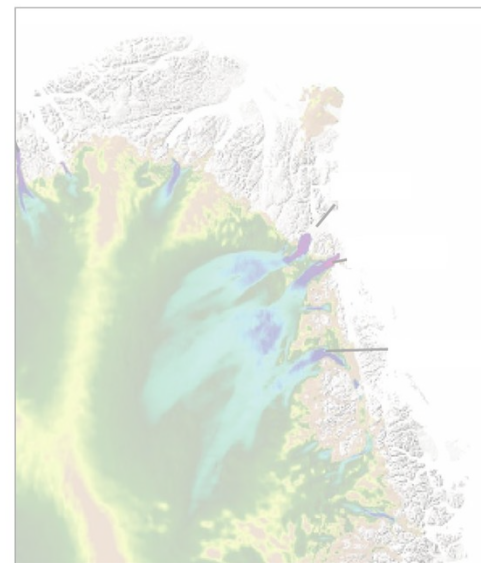
observed



Surface velocity [Aschwanden et al., 2016]



simulated
assuming viscous flow



Thanks for your attention!

Fiber-optic exploration of the cryosphere

Fichtner, Walter, Hudson et al., GJI 2026



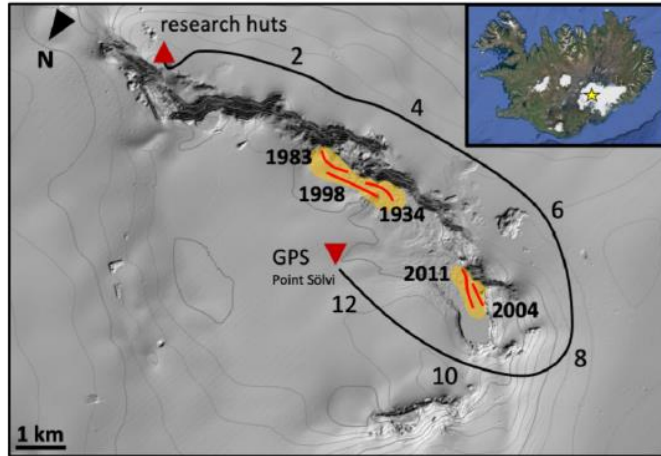
Addenda

Grimsvötn

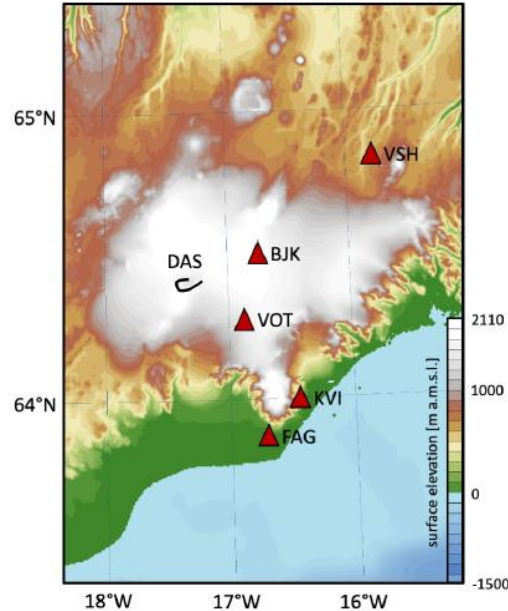
Setup

Experimental arrangement

a) Grimsvötn topography and DAS cable geometry



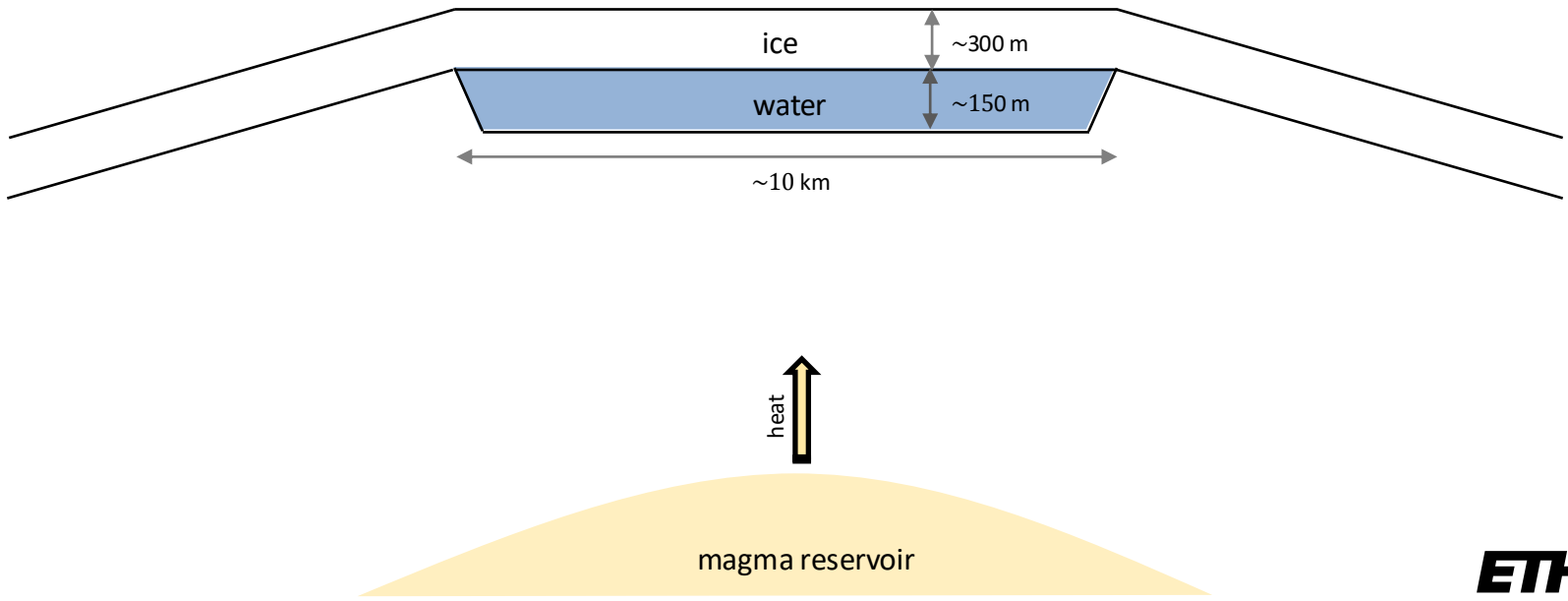
b) DAS and station map



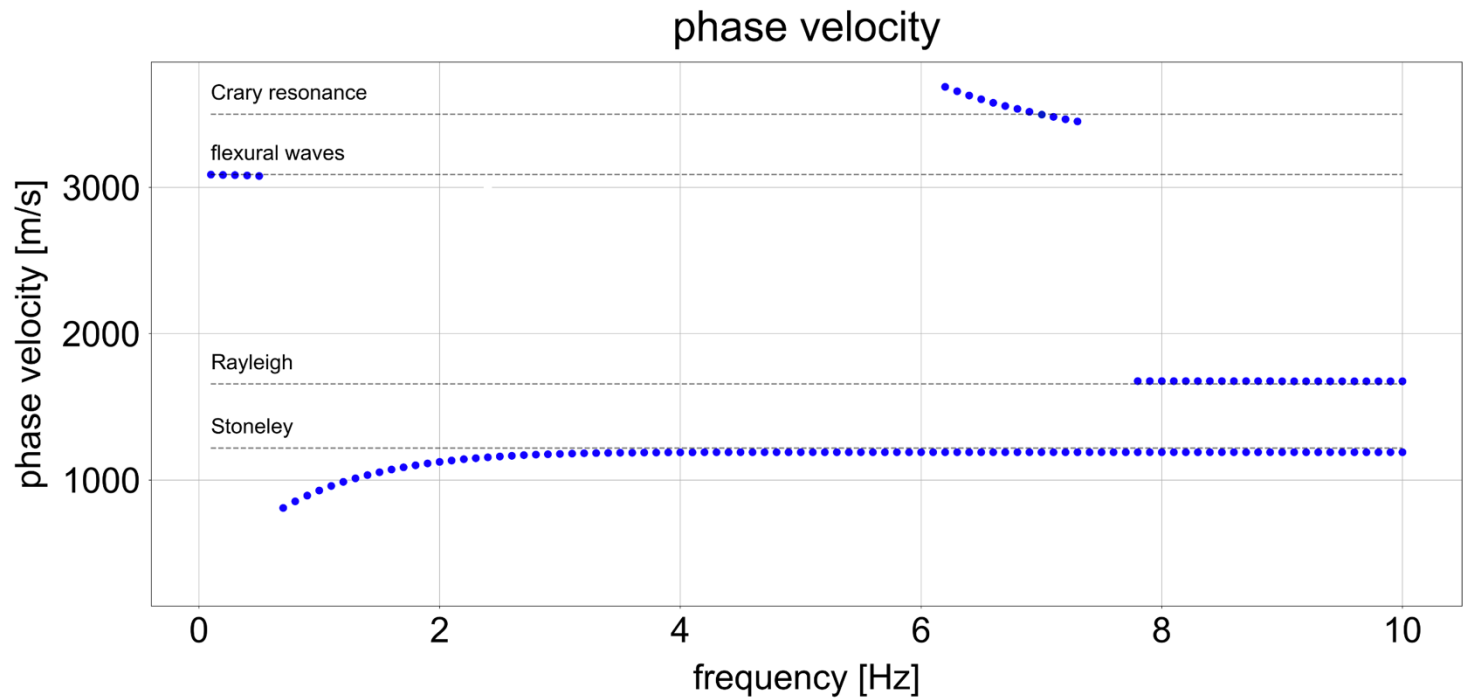
- Nearly 13 km of fibre-optic cable along the caldera of Grimsvötn.
- Recording of surface deformation during 3 weeks in May 2021.
- One recording every 8 m.

Setup

Schematic cross section

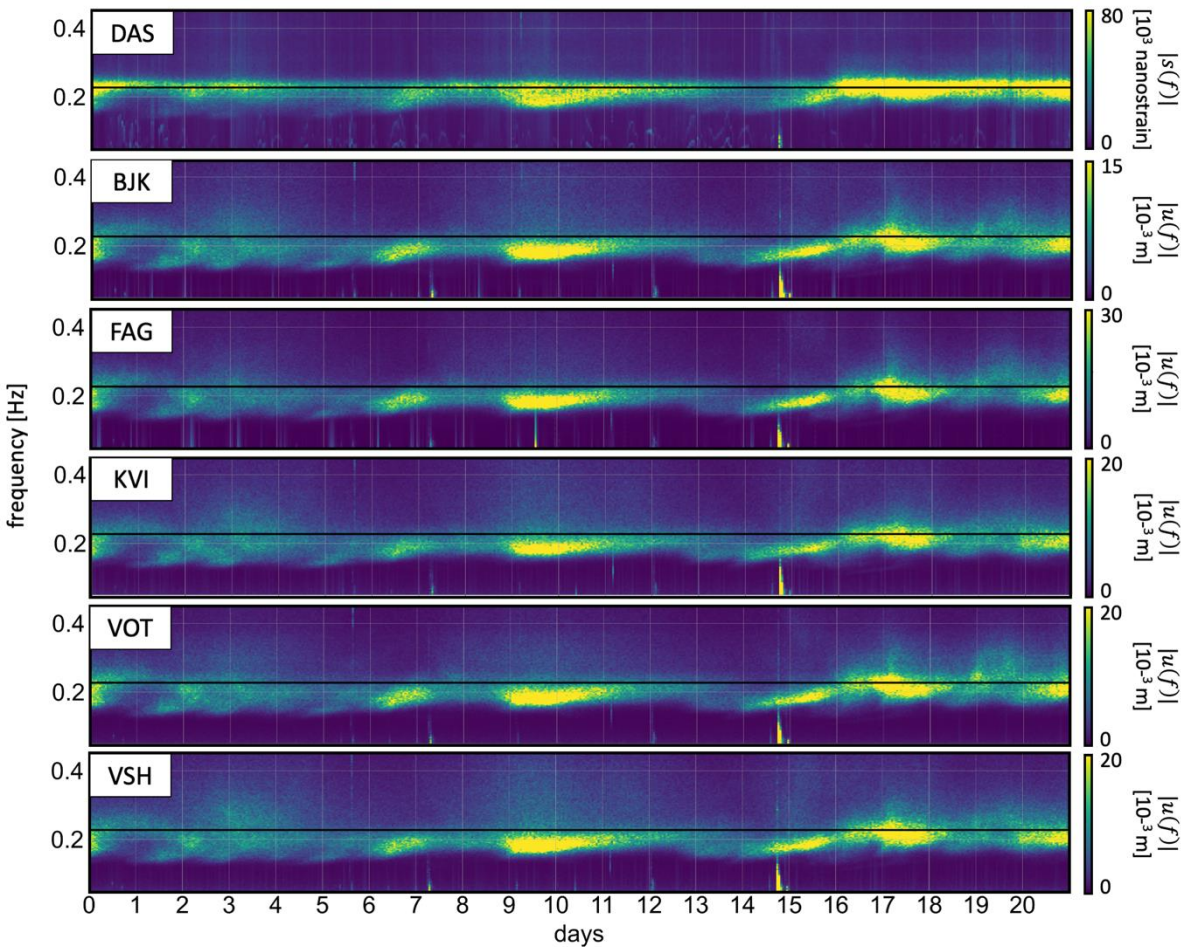


Waves in a floating ice sheet

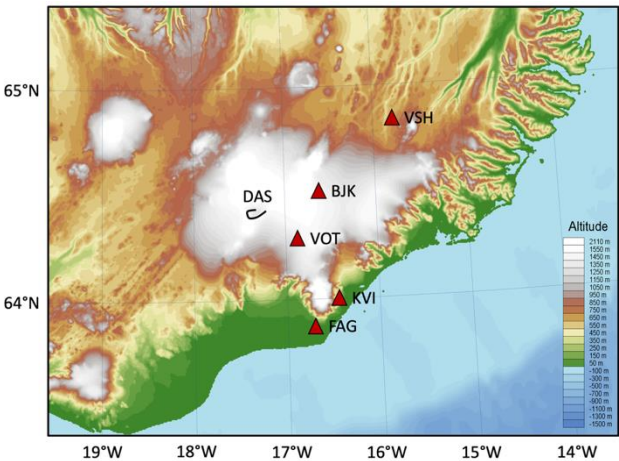


At frequencies around 0.2 Hz, all waves are flexural waves.

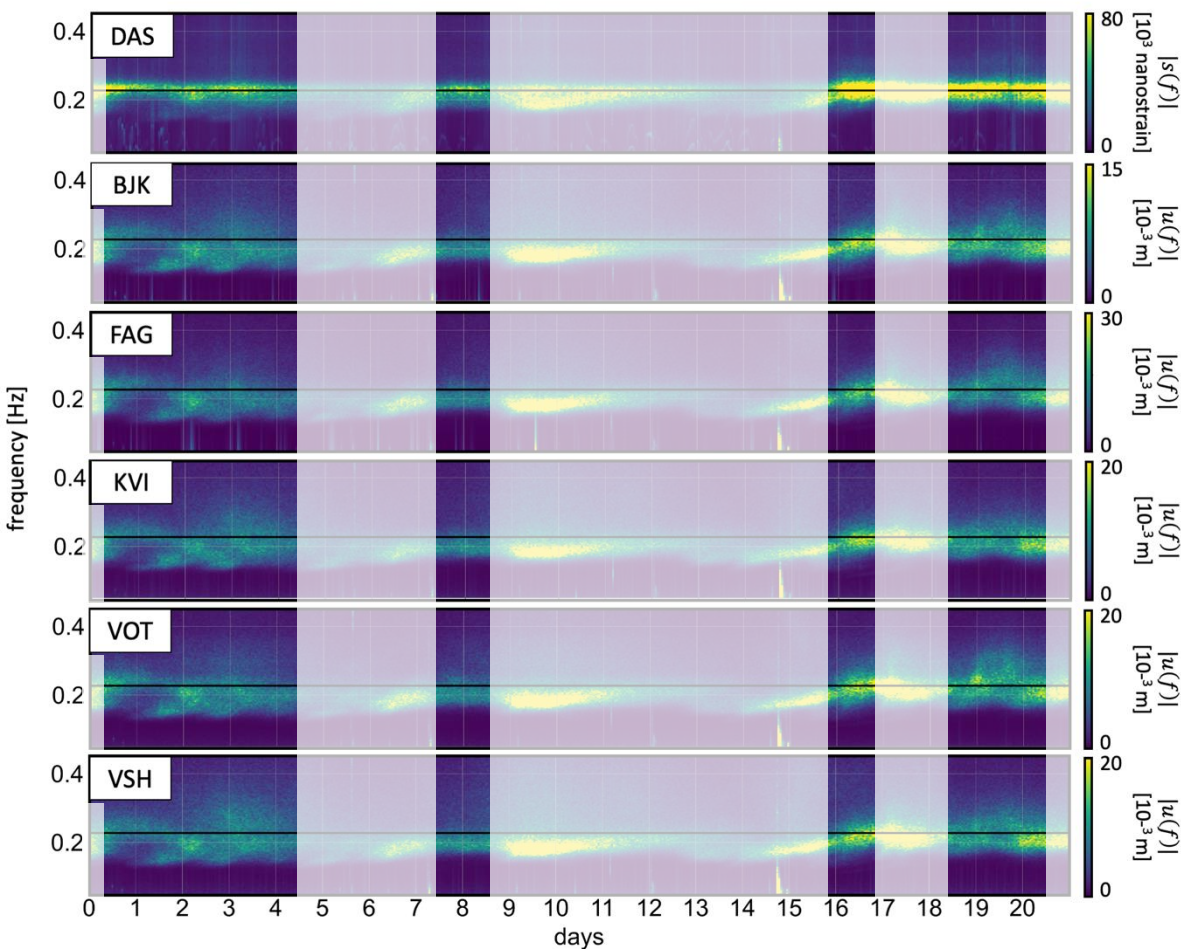
Spectrograms over 3 weeks



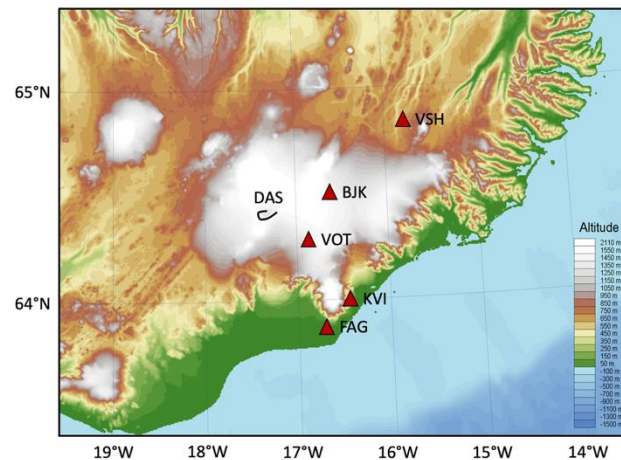
- Spectrograms of ambient vibrations at stations in southern Iceland are remarkably similar!
- Overall, they correlate with the spectrogram of the ice sheet oscillations.
- However ...



Spectrograms over 3 weeks



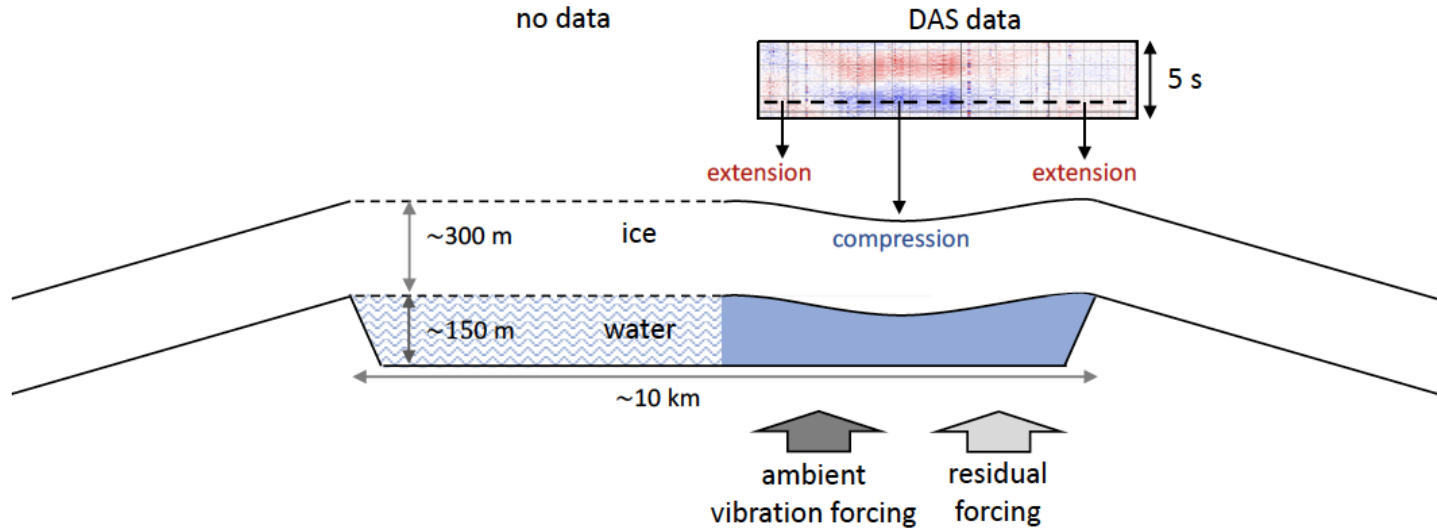
- Spectrograms of ambient vibrations at stations in southern Iceland are remarkably similar!
- Overall, they correlate with the spectrogram of the ice sheet oscillations.
- However, there are exceptions, as seen before:



A simple model

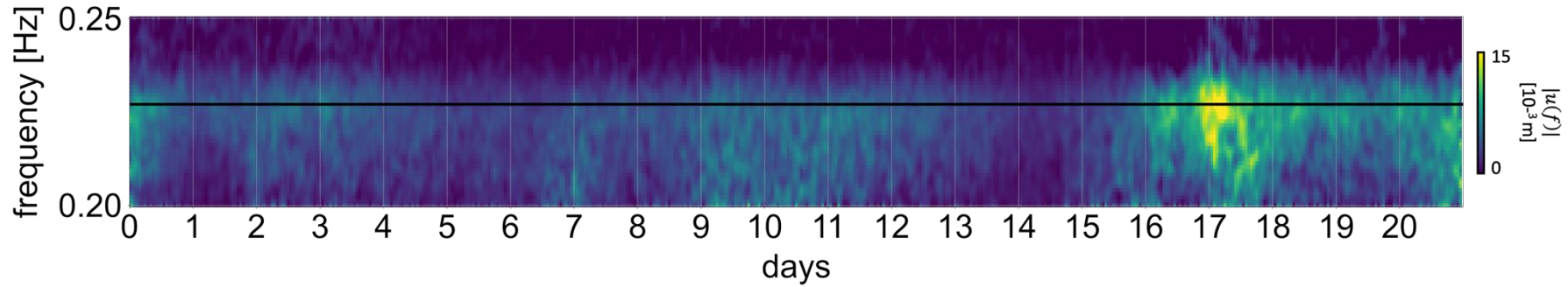
Ice sheet oscillations can only partly be explained by ambient vibration forcing.

What are the minimum forces required to explain the observations?



Minimum required forces

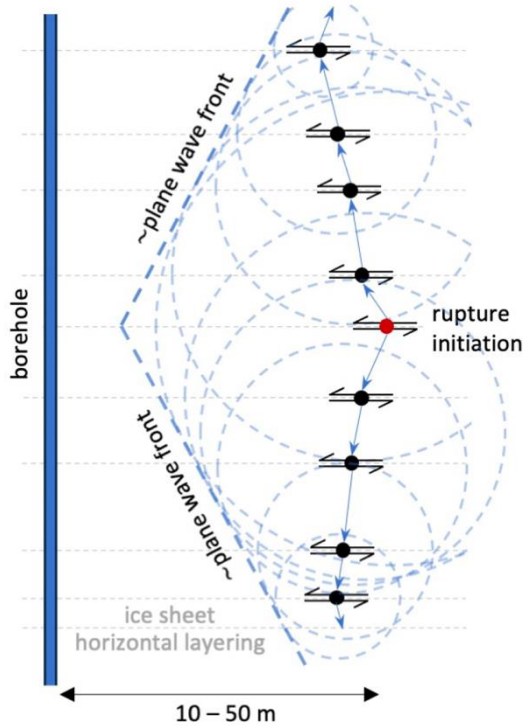
Using methods from Inverse Theory, we can estimate $f_{volcanic}$.



NEGIS

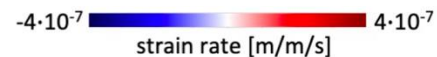
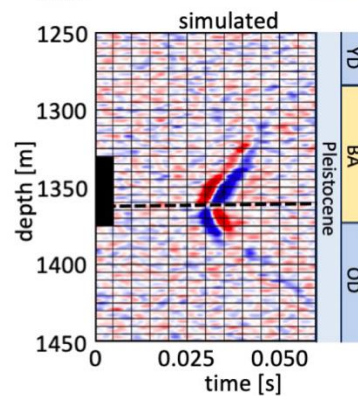
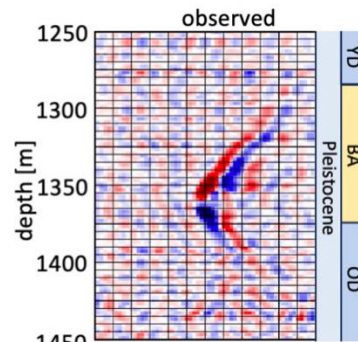
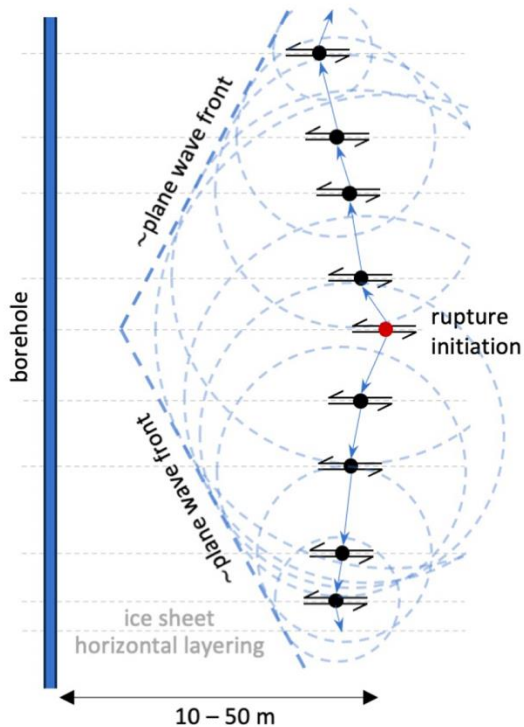
MODELLING VERTICAL EVENT CASCADES

a) schematic rupture mechanism



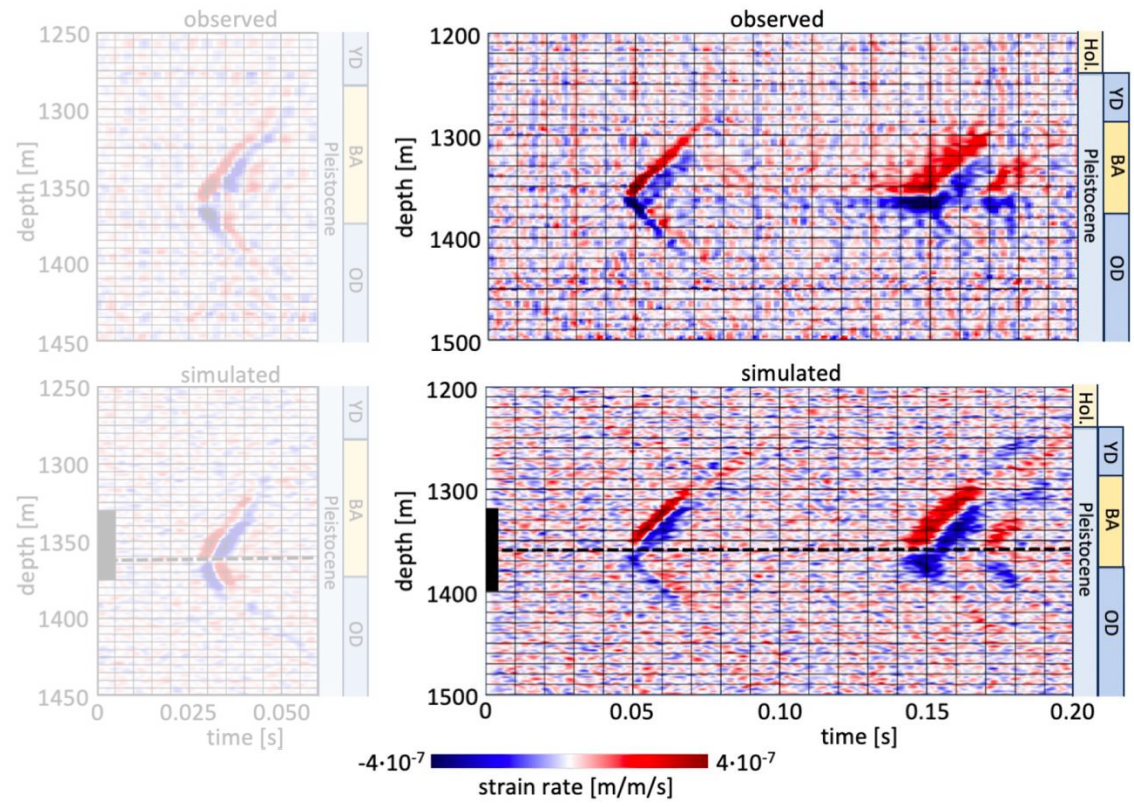
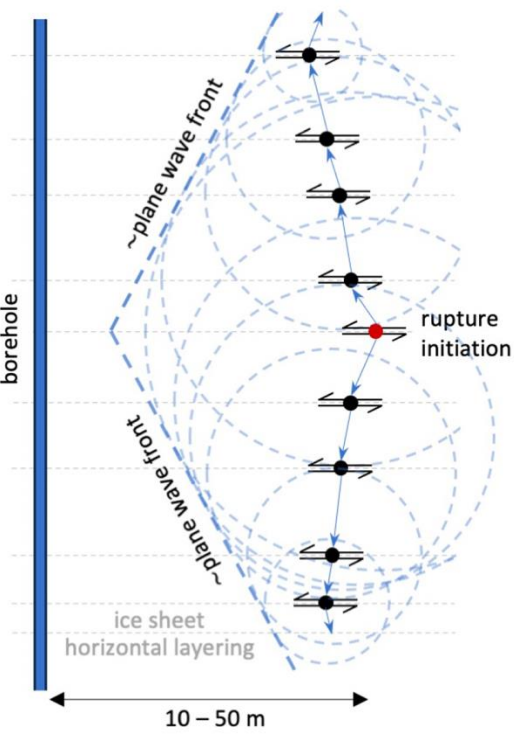
MODELLING VERTICAL EVENT CASCADES

a) schematic rupture mechanism

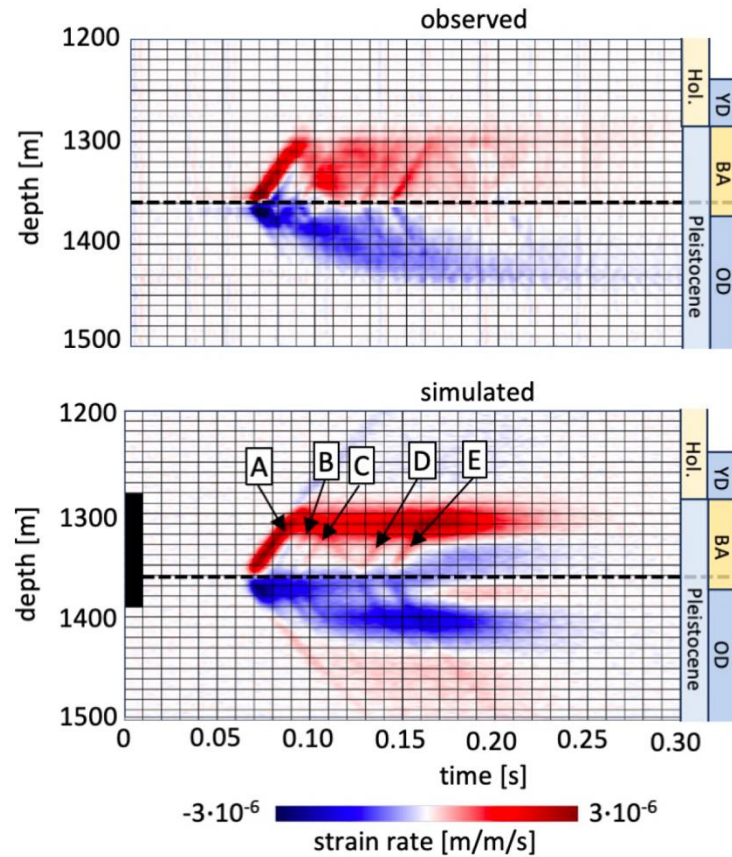


MODELLING VERTICAL EVENT CASCADES

a) schematic rupture mechanism



MODELLING VERTICAL EVENT CASCADES



MODELLING VERTICAL EVENT CASCADES

